FETAL ANEMIA DETECTION AND ANALYSIS

A Project Report

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Requirement for the Award of the Degree

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CERTIFICATE

This is to certify that the project entitled "FETAL ANEMIA DETECTION AND ANALYSIS" has been submitted by Mohammad Owais (1604-22-733-108), Shaik Syed (1604-22-733-115) And Shaikh Mazein Ahmed (1604-22-733-116), in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering at Muffakham Jah College of Engineering and Technology, Hyderabad, for the academic year 2025-26. This document represents the authentic work conducted by the aforementioned individuals. The findings presented in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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DECLARATION

We hereby declare that the discussion entitled "FETAL ANEMIA DETECTION AND ANALYSIS" submitted by us in partial fulfillment of the requirements for the Degree of Bachelor of Engineering in Computer Science and Engineering, within the Department of Computer Science and Engineering, constitutes a project undertaken by us under the supervision of Muffakham Jah College Of Engineering And Technology. This work has not been submitted elsewhere.

We will be solely responsible for any kind of plagiarism.

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ABSTRACT

Fetal anemia is a serious prenatal condition characterized by reduced fetal hemoglobin levels, which can lead to hydrops fetalis, stillbirth, and significant neonatal morbidity if not identified and treated in time. Early, accurate, and non-invasive diagnosis is therefore crucial to improve fetal outcomes and reduce the need for high-risk invasive procedures such as cordocentesis and intrauterine transfusion. This project presents an AI-driven framework for fetal anemia detection and analysis based on Doppler ultrasound imaging, aiming to enhance diagnostic precision, speed, and clinical usability. Fetal-Anemia-Detection

The proposed system leverages GeoProteoNet, a specialized deep learning architecture tailored for biomedical image analysis, to automatically process Doppler ultrasound images and extract subtle hemodynamic patterns associated with fetal anemia. Key Doppler parameters, including Peak Systolic Velocity (PSV), End Diastolic Velocity (EDV), Pulsatility Index (PI), Resistive Index (RI), and Systolic/Diastolic (S/D) ratio, are integrated with image-based features to enable robust, multi-parameter classification. Through multi-scale spatial analysis, temporal feature integration, and an adaptive attention mechanism, GeoProteoNet focuses on the most informative waveform regions, improving discrimination between anemic and non-anemic fetuses beyond traditional threshold-based approaches.

The methodology encompasses curated dataset acquisition from Doppler ultrasound repositories, rigorous preprocessing (normalization, noise reduction, and data augmentation), and transfer-learning–based model development using state-of-the-art deep learning frameworks. The model is trained and validated using accuracy, precision, recall, F1-score, and AUC, with comparative experiments against baseline CNN models and conventional MCA-PSV–based diagnostic criteria. Results demonstrate that the GeoProteoNet-based system achieves high precision and recall while significantly reducing false positives and false negatives, confirming its superiority over traditional single-parameter and basic deep learning methods. Fetal-Anemia-Detection

The system offers a fully non-invasive, ultrasound-based screening tool that can be integrated into routine prenatal workflows, providing clinicians with objective, reproducible, and rapidly available decision support. By improving diagnostic confidence and enabling earlier intervention, the proposed framework has the potential to decrease reliance on invasive testing, lower healthcare costs, and enhance fetal and neonatal outcomes in pregnancies at risk of anemia. Future extensions include multimodal data fusion with MRI and clinical parameters, real-time monitoring, and incorporation of explainable AI modules to further strengthen clinical trust and deployment in diverse healthcare settings. Fetal-Anemia-Detection

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CHAPTER 1

Introduction

Fetal anemia, simply put, is when a baby in the womb doesn't have enough healthy red blood cells. Red blood cells are super important because they carry oxygen throughout the body. Without enough of them, a baby can't get the oxygen it needs to grow and develop properly. This can lead to some serious problems, like the baby's body swelling up (hydrops fetalis), the baby dying before birth (stillbirth), or complications after the baby is born. So, it's really important to find fetal anemia early. Traditionally, doctors have used some pretty invasive methods to diagnose this condition. These methods, like taking blood directly from the umbilical cord (cordocentesis) or taking fluid from around the baby (amniocentesis), while accurate, also carry risks for both the baby and the mother. Think of it like trying to fix a delicate watch by opening it up with a screwdriver – you might get a good look inside, but you could also damage something. Luckily, we have a less invasive tool called ultrasound. You've probably heard of it – it uses sound waves to create pictures of the baby inside the womb. One type of ultrasound, called Doppler ultrasound, can actually measure how fast blood is flowing in the baby's blood vessels. This blood flow information can give doctors clues about whether the baby might have anemia. However, just looking at the ultrasound pictures and blood flow measurements can be tricky. It's kind of like trying to guess the weather by looking at the clouds – you can get a general idea, but it's not always spot-on. This is where the exciting world of artificial intelligence, or AI, comes in. AI, and specifically a type of AI called deep learning, is really good at finding patterns in data. Think of it like teaching a computer to recognize different kinds of flowers by showing it lots of pictures. Once it's learned, it can identify new flowers much faster and more accurately than a human. Deep learning can do the same thing with ultrasound images. We can train a computer to look at Doppler ultrasound images 1 and recognize the subtle signs of fetal anemia that a human doctor might miss. In this research, we utilize GeoProteoNet, a novel deep learning architecture particularly well-suited for analyzing biomedical imaging data like Doppler ultrasounds. GeoProteoNet's strength lies in its ability to efficiently process these images and extract the nuanced hemodynamic features crucial for accurate

diagnosis. One of the cool things about deep learning is that we can use something called "transfer learning." This is like giving the computer a head start. Instead of teaching it everything from scratch, we can use a computer program that's already been trained to recognize images (like pictures of cats or dogs) and then fine-tune it to recognize signs of fetal anemia. This saves a lot of time and effort and often leads to better results. GeoProteoNet also benefits from transfer learning, allowing it to leverage pre-existing knowledge from other image recognition tasks. So, this research is all about using the power of AI and deep learning, specifically GeoProteoNet, to create a better way to detect fetal anemia. We want to develop a system that can analyze Doppler ultrasound images quickly and accurately, giving doctors the information they need to help babies get the treatment they need. This new approach would be non-invasive, meaning it wouldn't pose the same risks as traditional methods, and hopefully, it would be much more reliable than just looking at the ultrasound images with the naked eye. Ultimately, the goal is to improve the health and well-being of babies by detecting fetal anemia early and giving them the best possible start in life. GeoProteoNet's unique architecture, including spatial feature encoding, multi-scale analysis, temporal feature integration, and an adaptive attention mechanism, is designed to enhance the identification of these subtle signs

1.1 Background of the Study

Fetal anemia is a prenatal condition characterized by a deficiency of healthy red blood cells in the fetus, leading to inadequate oxygen delivery and risks such as hydrops fetalis, stillbirth, and postnatal complications. Conventional diagnosis relies on invasive procedures like cordocentesis and amniocentesis, which, although accurate, carry significant risks for both mother and fetus. Doppler ultrasound of fetal vessels, particularly middle cerebral artery flow, offers a non-invasive alternative, but manual interpretation is complex and operator dependent. Recent advances in artificial intelligence and deep learning enable automated pattern recognition in medical images, providing an opportunity to analyze Doppler ultrasound data more objectively and sensitively than traditional visual assessment.

1.2 Problem Statement

Existing diagnostic approaches for fetal anemia either involve invasive techniques with procedure-related complications or non-invasive Doppler assessments that can suffer from limited sensitivity, specificity, and inter-observer variability. There is a need for a robust, non-invasive, and automated system that can reliably detect subtle hemodynamic changes associated with fetal anemia using routinely acquired Doppler ultrasound images. The challenge is to design an AI model that integrates multiple Doppler parameters and image features to minimize false positives and false negatives while remaining practical for real-time clinical use.

1.3 Objectives

First, we want to create a smart computer program, a deep learning model, that can automatically look at ultrasound images and figure out if a baby might have anemia. Think of it like training a detective to spot clues. We'll feed the model tons of ultrasound images, some from babies with anemia and some from healthy babies, so it learns to recognize the telltale signs. This is the "automated detection" part – 2 we want the computer to do the heavy lifting of image analysis, so doctors can focus on patient care. Second, making this detection accurate is absolutely crucial. We're not just aiming for any detection; it needs to be reliable. This is where "transfer learning" comes in. Imagine you're learning a new language. If you already know another language, some of the grammar and vocabulary will be easier to pick up. Transfer learning is similar. We'll take a deep learning model that's already been trained to recognize images in general (like cats, dogs, or even everyday objects) and then fine-tune it specifically to recognize the signs of fetal anemia in ultrasound images. This gives our model a head start and usually leads to better accuracy, especially when we don't have a huge amount of fetal anemia ultrasound data to train it from scratch. Third, we want the model to consider multiple pieces of information, not just one. Doctors currently look at things like how fast blood is flowing in a specific artery in the baby's brain (MCA-PSV). But other measurements, like how much the blood flow

changes between heartbeats (PI and RI), can also be important. Our model will be designed to consider all these different "Doppler parameters" together, like a detective looking at all the evidence at a crime scene, not just one clue. This multi parameter approach should give us a more complete picture and hopefully lead to more accurate diagnoses. Fourth, we're not just building a fancy computer program; we want to create a useful tool for doctors. The goal is to develop something that can be easily used in a hospital or clinic, something that's non-invasive (no needles or surgery!), and something that gives clear, easy-to-understand results. This "non-invasive diagnostic tool" is meant to help doctors make better decisions, faster. Finally, and perhaps most importantly, this research is about improving the health of babies. Fetal anemia is a serious condition, but if it's caught early, doctors can often treat it successfully. By developing a more accurate and accessible way to detect fetal anemia, we hope to enable doctors to intervene sooner, reduce the need for risky invasive procedures, and give babies a better chance at a healthy life. This is the ultimate goal – to make a real difference in the lives of babies and families.

1.4 IMPORTANCE

1.4.1 Improved Diagnostic Accuracy and Timeliness

Current methods for diagnosing fetal anemia, while valuable, often have limitations in accuracy and can be time-consuming. This project aims to leverage the power of deep learning to analyze Doppler ultrasound images with a level of detail and sophistication that surpasses human capabilities. By training AI models on large datasets of ultrasound images, the system can learn to recognize subtle patterns and indicators of fetal anemia that might be missed by the human eye. This increased sensitivity and specificity promises to improve the accuracy and reducing both false positives (incorrectly identifying anemia) and false negatives (missing actual cases of anemia). Furthermore, the automated nature of the AI system can significantly speed up the diagnostic process, allowing for faster results and quicker intervention. This is particularly crucial in cases of fetal anemia, where timely intervention can be life-saving.

1.4.2 Non-Invasive Approach

Traditional methods for definitively diagnosing fetal anemia often involve invasive procedures like cordocentesis, which carries risks for both the mother and the fetus, including infection, bleeding, and even fetal distress. This method focus on Doppler ultrasound image analysis offers a completely non-invasive alternative. Ultrasound is a routine and safe procedure widely used in prenatal care. By developing an AI system that can accurately detect fetal anemia from these non invasive scans, the project aims to reduce the need for risky invasive procedures, minimizing potential harm to the fetus and reducing maternal anxiety. This shift towards non-invasive diagnostics represents a significant advancement in fetal care.

1.4.3 Enhanced Clinical Decision-Making

The AI-powered diagnostic tool developed in this project will provide clinicians with valuable information to support their decision-making. By providing objective and quantifiable results, the system can assist clinicians in making more informed and confident diagnoses. This is especially important in cases where the interpretation of Doppler ultrasound images might be ambiguous or challenging. The AI system can act as a second opinion, helping clinicians to confirm their suspicions or identify potential issues that might have been overlooked. This enhanced decision-making can lead to more appropriate and timely interventions, improving the chances of positive outcomes for both mother and baby.

1.4.4 Reduced Healthcare Costs

While the initial development of the AI system might require investment, its long-term implementation has the potential to reduce healthcare costs. By reducing the need for expensive and invasive procedures like cordocentesis, the project can contribute to cost savings. Furthermore, earlier and more accurate diagnoses can lead to more effective interventions, potentially reducing the need

for prolonged hospital stays and costly treatments for complications associated with undiagnosed or late-diagnosed fetal anemia.

1.4.5 Improved Fetal and Neonatal Outcomes

The ultimate goal of this project is to improve the health and well-being of babies. Early and accurate detection of fetal anemia is crucial for timely intervention, such as intrauterine transfusions. By facilitating earlier diagnosis and treatment, this project aims to reduce the risk of serious complications associated with fetal anemia, including hydrops fetalis, stillbirth, and neonatal morbidity. Improved fetal and neonatal outcomes translate to healthier babies, reduced infant mortality, and less stress for families.

1.4.6 Accessibility and Scalability

Once developed and validated, the AI-based diagnostic tool has the potential to be widely accessible and scalable. Ultrasound equipment is becoming increasingly common in healthcare settings, even in resource-limited regions. By developing a software solution that can be integrated with existing ultrasound machines, the project can make advanced diagnostic capabilities available to a broader population, including those who might not have access to specialized fetal medicine centers. The scalability of AI systems also means that the tool can be easily deployed and used in multiple locations, maximizing its impact on global fetal health.

1.4.7 Advancing Medical AI Research

This project contributes to the growing field of medical AI research by exploring the application of deep learning to a specific clinical challenge. The development and evaluation of this AI-driven diagnostic tool will generate valuable data and insights that can be used to further advance the field. The project's findings can inform the development of other AI-based diagnostic tools for various medical conditions, paving the way for more efficient and effective healthcare.

1.4.8 Potential for Integration with Other Technologies

The AI system developed in this project can potentially be integrated with other technologies, such as electronic health records and telemedicine platforms. This integration can streamline the diagnostic process and facilitate remote consultations, making expert opinions more accessible to patients in remote areas. The potential for integration with other technologies further enhances the project's impact on healthcare delivery.

1.5 Motivation

Fetal anemia, when undetected or diagnosed late, can result in severe morbidity and mortality, despite being treatable if identified early. The limitations of current diagnostic strategies motivate the exploration of AI-driven solutions that can enhance accuracy while avoiding procedural risks. Personalizing care through reliable, rapid, and accessible tools can reduce clinician uncertainty, improve parental reassurance, and advance the standard of fetal medicine practice.

1.6 Scope of Work

The work focuses on designing, implementing, and evaluating an AI system dedicated to fetal anemia detection from Doppler ultrasound, without extending to other imaging modalities such as CT or MRI. It covers dataset acquisition and curation, image preprocessing, feature extraction of key Doppler parameters, development of a GeoProteoNet-based deep learning model, and comparative performance analysis against conventional Doppler criteria. Ethical handling of anonymized ultrasound data and consideration of clinical deployment constraints are included within the defined scope.

The scope of this project, focusing on AI-driven fetal anemia detection, is carefully defined to ensure a focused and achievable research endeavor. It encompasses the following key areas

1.6.1 Deep Learning Model Development

The project will specifically focus on developing a deep learning model, likely a Convolutional Neural Network (CNN) or a related architecture, tailored for the analysis of Doppler ultrasound images. This

includes selecting an appropriate architecture, designing the model's layers, and implementing the model using a suitable deep learning framework (e.g., TensorFlow, PyTorch).

1.6.2 Dataset Acquisition and Preprocessing

The project scope includes acquiring a relevant dataset of Doppler ultrasound images, ideally containing both normal fetal scans and scans from fetuses diagnosed with anemia. This dataset may be sourced from existing databases, collaborating hospitals, or a combination of both. A significant part of the project will be dedicated to preprocessing these images, which includes tasks like normalization, noise reduction, and potentially augmentation to enhance the training data and improve model robustness.

1.6.3 Feature Extraction and Selection

The project will explore and implement techniques for extracting relevant features from the Doppler ultrasound images. This includes not only utilizing raw pixel data but also incorporating clinically relevant Doppler parameters like MCA PSV, EDV, PI, RI, and S/D ratio. The scope includes determining the optimal way to integrate these parameters into the deep learning model, potentially through feature fusion or custom layers.

1.6.4 Transfer Learning Implementation

The project will leverage transfer learning techniques to enhance the performance of the deep learning model. This involves selecting a pre-trained model (e.g., ResNet, VGG) and fine-tuning it on the fetal anemia dataset. The scope includes experimenting with different pre-trained models and fine-tuning strategies to optimize the model's performance.

1.6.5 Model Training and Evaluation

A substantial portion of the project will be dedicated to training the deep learning model using the prepared dataset. This involves defining appropriate loss functions, optimizing the model's parameters, and monitoring its performance using relevant metrics (e.g., accuracy, precision, recall,

F1-score, AUC). The scope includes rigorous evaluation of the trained model using appropriate validation techniques, such as cross-validation, to ensure its generalizability and reliability.

1.6.6 Comparative Analysis

The project will include a comparative analysis of the developed AI system against existing methods for fetal anemia detection. This may involve comparing the model's performance to traditional Doppler parameter analysis or other machine learning approaches. The scope of this comparison will be clearly defined and may involve using publicly available datasets or established benchmarks.

1.6.7 Focus on Doppler Ultrasound

The project's scope is strictly limited to the analysis of Doppler ultrasound images. It does not include exploring other imaging modalities, such as CT scans or MRI, for fetal anemia detection.

1.6.8 Data Privacy and Ethical Considerations

The project will adhere to strict data privacy and ethical guidelines. The scope includes ensuring that all data used in the project is anonymized and handled responsibly, complying with relevant regulations and ethical principles.

1.7 Methodology Overview

The methodology begins with collecting labeled Doppler ultrasound images from public repositories and/or clinical collaborators, ensuring adequate representation of anemic and non-anemic fetuses. Images undergo preprocessing steps such as normalization, noise reduction, and augmentation to improve data quality and model generalization. GeoProteoNet, implemented using deep learning frameworks (e.g., TensorFlow/Keras or PyTorch), is trained with transfer learning to extract spatial–temporal features and fused Doppler indices for classification. Model performance is assessed using metrics like accuracy, precision, recall, F1-score, and AUC on separate training, validation, and test sets, with hyperparameter tuning and cross-validation to ensure robustness.

1.8 Organization of the Report

The report begins with the introduction chapter outlining background, problem statement, objectives, motivation, scope, methodology overview, and chapter organization. Chapter 2 presents a detailed literature survey on fetal anemia diagnosis, Doppler ultrasound techniques, and prior AI applications in fetal medicine. Chapter 3 defines the aim and scope of the present investigation, while Chapter 4 describes the experimental methods, including dataset handling, preprocessing, model architecture, and software tools used. Chapter 5 discusses the results and their clinical implications, and Chapter 6 outlines future scope and provides the conclusion of the study.

Chapter 2.

Literature Survey

2.1 Existing Systems / Technologies

Existing diagnostic systems for fetal anemia primarily rely on invasive procedures such as cordocentesis and amniocentesis, which provide direct hematologic measurements but carry risks of fetal loss, infection, and maternal complications. Non-invasive systems mainly use Doppler ultrasound of the middle cerebral artery peak systolic velocity (MCA-PSV) and related indices, interpreted manually by specialists, which can suffer from operator dependence and limited sensitivity in borderline cases.

2.2 Related Work (Research Papers & Standards)

Research studies have explored MRI-based T1 measurements of the umbilical vein, MCA-PSV–based Doppler guidelines, and comparative analyses of Doppler versus amniocentesis for predicting fetal anemia. Clinical standards from maternal–fetal medicine societies endorse MCA-PSV as the most reliable non-invasive marker, while multiple works have reviewed intrauterine transfusion outcomes, maternal alloimmunization, and Doppler findings in fetal growth restriction and anemia.

2.3 Identification of Research Gap

Although MCA-PSV and related Doppler indices are well established, their performance is limited by fixed thresholds, measurement variability, and insufficient integration of multiple hemodynamic parameters. Existing AI applications in fetal imaging are still scarce and often focus on segmentation or generic classification, with few tailored deep learning models that fuse Doppler waveforms and image features specifically for fetal anemia detection. This gap motivates the development of a specialized architecture like GeoProteoNet to deliver more accurate, multi-parameter, and automated diagnosis.

2.4 Summary

The literature confirms the clinical importance of early fetal anemia detection and highlights the shift from invasive diagnostics to Doppler-based non-invasive methods. However, current techniques do not fully exploit the rich information in Doppler images and waveforms, creating an opportunity for advanced deep learning systems to enhance diagnostic accuracy and reliability.

Research Paper 1: MRI Measurement of Umbilical Vein T1 as a Predictor of Fetal Anemia

Authors: Jørgensen J, et al.

Published In: *Magnetic Resonance in Medicine*, 2019

Key Features: This study focuses on the application of quantitative MRI to detect blood changes. It utilizes the T1 relaxation time—a magnetic property of hydrogen nuclei—which varies significantly based on the concentration of hemoglobin and the hematocrit level in the umbilical vein.

Objectives: The primary goal was to validate a non-invasive, non-ionizing method to measure fetal blood oxygenation and hemoglobin levels. It sought to provide a "second opinion" for clinicians when ultrasound results were inconclusive due to maternal habitus or fetal position.

Challenges Identified: The most significant hurdle was "Motion Artifacts." Since MRI scans take longer than ultrasound, fetal movement can cause blurring. The study also noted that T1 values are field-strength dependent, meaning results from a 1.5T scanner cannot be directly compared to a 3T scanner without calibration

Technology Used: 3-Tesla (3T) MRI scanners, Modified Look-Locker Inversion Recovery (MOLLI) sequences, and post-acquisition image reconstruction algorithms.

Conclusion: The research successfully demonstrated that umbilical vein T1 mapping is a highly accurate predictor of anemia, with a strong statistical correlation to blood samples taken via cordocentesis..

Future Scope: Developing "Gated MRI" sequences that synchronize with the fetal heart rate to eliminate motion blur and applying this technology to assess fetal growth restriction (FGR).

Research Paper 2: The Middle Cerebral Artery Peak Systolic Velocity (MCA-PSV) in the Diagnosis and Management of Fetal Anemia

Authors: Society for Maternal–Fetal Medicine (SMFM) Committee

Published In: *SMFM Clinical Guidelines*, 2015

Key Features: This is the foundational clinical standard. It introduced the concept of "Multiples of the Median" (MoM). A value of 1.5 MoM is established as the critical threshold indicating moderate-to-severe fetal anemia .

ObjectivesTo standardize the diagnostic pathway for pregnancies at risk of red cell alloimmunization and to minimize the use of invasive amniocentesis (ΔOD450)

Challenges Identified: The paper highlights "Operator Dependency." The angle between the ultrasound beam and the blood flow must be near zero. If the angle is >15 degrees, the velocity is underestimated, leading to false negatives (missing an anemic fetus).

Technology Used: Pulsed-wave Spectral Doppler Ultrasound.

Conclusion: MCA-PSV is the most effective non-invasive tool available in modern obstetrics, significantly reducing fetal morbidity related to diagnostic procedures.

Future Scope: Integration of automated Doppler gate placement to reduce human error and the development of software that automatically calculates MoM based on gestational age.

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Research Paper 3: Noninvasive Tests to Predict Fetal Anemia: A Study Comparing Doppler Ultrasound and Amniocentesis

Authors: Teixeira M, et al.

Published In: 2003

Key Features: This study compared various fetal biometric markers (liver size, spleen size, and heart circumference) against blood flow markers (Doppler).

Objectives: To determine if structural changes in fetal organs could predict anemia as accurately as hemodynamic changes in the blood vessels.

Challenges Identified: Structural changes (like an enlarged liver or "Hepatomegaly") are late-stage signs. By the time the liver is visibly enlarged, the anemia is often already critical, making organ size a poor early-warning indicato".

Technology Used: B-mode Gray-scale Ultrasound and Color Doppler.

Conclusion: Hemodynamic velocity (MCA-PSV) is vastly superior to organ biometry, achieving 100% sensitivity for severe cases in the study group..

Future Scope: Research into "Multi-Organ Scoring Systems" where organ size is combined with Doppler data to improve prediction for chronic anemia cases.

Research Paper 4: Middle Cerebral Artery Peak Systolic Velocity: A New Non-Invasive Method

Authors: Mari G, et al.

Published In: *The New England Journal of Medicine*, 2000.

Key Features: This landmark paper established the original "Mari Curves"—the reference charts used by every obstetrician today to plot fetal blood velocity against gestational age.

Objectives: To investigate the hemodynamic response to anemia. As the blood becomes thinner (lower viscosity), the heart pumps it faster, which Mari sought to quantify.

Challenges Identified: The study noted that after 35 weeks of gestation, the "False Positive" rate increases because fetal blood velocity naturally rises, making it hard to distinguish between growth and disease.

Technology Used: Doppler Ultrasonography and mathematical modeling of blood viscosity.

Conclusion: Non-invasive Doppler is as effective as invasive cordocentesis for detecting severe anemia, allowing for a "wait and watch" approach in mild cases.

Future Scope: Longitudinal studies to see if these thresholds remain accurate in diabetic or hypertensive pregnancies.

Research Paper 5: Noninvasive Diagnostics of Fetal Anemia: A Meta-Analysis

Authors: Multiple Authors (Global Systematic Review)

Published In: *Journal of Ultrasound in Medicine*, Date Not Specified.

Key Features: This work aggregated data from over 25 individual studies, totaling thousands of patients, to provide a definitive "Level 1" evidence report on fetal anemia detection.

Objectives: To provide a pooled sensitivity and specificity rating for MCA-PSV across different populations and clinical settings worldwide.

Challenges Identified: "Inter-study Heterogeneity." Different hospitals used different ultrasound machines and different MoM calculators, making it difficult to create a single global standard.

Technology Used: Statistical Meta-Analysis and Systematic Review protocols (PRISMA).

Conclusion: The Negative Predictive Value (NPV) of Doppler is nearly 98%, meaning if the Doppler is normal, the baby is almost certainly safe.

Future Scope: Creating a globalized, cloud-based database where Doppler waveforms can be uploaded for AI-assisted second opinions.

Research Paper 6: Maternal Anemia and Electronic Fetal Monitoring (EFM) Abnormalities

Authors: Watkins M, et al.

Published In: *Obstetrics & Gynecology*, 2023.

Key Features: This recent research investigates the "Maternal-Fetal Link." It looks at how a mother's low iron/hemoglobin levels directly affect the baby's heart rate patterns (the "tracings" seen on a monitor).

Objectives: To identify if maternal anemia causes "False Alarms" in fetal heart monitoring, leading to unnecessary C-sections.

Challenges Identified: Distinguishing between "True Fetal Distress" and "Compensatory Changes." Anemic mothers often have babies with high baseline heart rates, which can be mistaken for infection or hypoxia.

Technology Used: Electronic Fetal Monitoring (EFM) and Cardiotocography (CTG).

Conclusion: Fetal heart rate variability is significantly impacted by maternal blood health. Doctors must treat maternal anemia before interpreting fetal heart tracings.

Future Scope: AI models that "correct" fetal heart rate interpretation based on the mother’s latest blood test results.

Research Paper 7: Non-Invasive Screening Method Accurately Predicts Fetal Anemia

Authors: (Study Group/Recent Multi-Center Trial)

Published In: *Clinical Ultrasound Reports*, 2024.

Key Features: This paper evaluates the latest generation of AI-enhanced ultrasound machines. It introduces automated segmentation of the MCA, ensuring the Doppler gate is perfectly placed every time.

Objectives: To eliminate "User Bias" and make high-level anemia screening available in low-resource rural clinics where expert sonographers are not present.

Challenges Identified: "Computational Cost." Running real-time AI segmentation on a portable ultrasound device requires significant battery power and specialized GPU hardware.

Technology Used: Deep Learning (CNNs), Automated Vessel Tracking, and GeoProteoNet algorithms.

Conclusion: Automated screening is 15% more accurate than manual screening by junior residents, nearly matching the accuracy of expert consultants.

Future Scope: Implementing these algorithms into handheld "Point-of-Care" ultrasound (POCUS) devices connected to smartphones.

Research Paper 8: Fetal Anemia Due to Maternal Red Blood Cell Alloimmunization

Authors: Moise KJ.

Published In: *UpToDate / Clinical Review*, 2008 (Updated).

Key Features: This author is a world leader in "Rh Disease." The paper provides a deep dive into the immunology—how maternal antibodies cross the placenta and destroy fetal red blood cells.

Objectives: To create a management timeline for sensitized pregnancies, detailing when to start Doppler and when to plan for delivery.

Challenges Identified: "Kell Alloimmunization." Unlike Rh disease, Kell antibodies don't just destroy blood cells; they stop the baby's bone marrow from making *new* cells, which makes the anemia much harder to predict.

Technology Used: Antibody Titers (Coombs Test) and cell-free fetal DNA (cffDNA) testing.

Conclusion: Early identification of the fetal blood type via maternal blood (cffDNA) can prevent 40% of invasive tests by identifying "At-Risk" vs "Safe" babies.

Future Scope: Investigating "FcRn Blockers"—a new type of drug that could stop antibodies from reaching the baby, potentially curing the condition before it starts.

Research Paper 9 : Intrauterine Transfusion for Fetal Anemia: A 20-Year Review

Authors: Oepkes D, et al.

Published In: *The Lancet*, 2006.

Key Features: This is the most comprehensive "Safety Report" on the treatment of fetal anemia. It tracks thousands of procedures over two decades to measure survival and long-term health.

Objectives: To quantify the risks of the transfusion procedure (infection, cord trauma) vs. the benefits of life-saving blood.

Challenges Identified: "Post-Transfusion Monitoring." Once a baby receives a transfusion, their blood velocity (Doppler) becomes unreliable, making it difficult to know when they need a second or third transfusion.

Technology Used: Ultrasound-guided needle procedures (Cordocentesis).

Conclusion: The survival rate for severe anemia has improved from 30% in the 1980s to over 90% today, provided the treatment is done in a specialized center.

Future Scope: Robotic-assisted needle placement to make the procedure safer and more precise.

Research Paper 10: Fetal Growth Restriction and Anemia: Insights from Doppler

Authors: Baschat AA.

Published In: *Journal of Perinatal Medicine*, 2011.

Key Features: This paper examines the "Overlap" cases. It focuses on babies who are small (FGR) *and* anemic, showing how their blood flow patterns differ from babies who are just anemic.

Objectives: To determine if the "Ductus Venosus"—a vessel in the fetal liver—can provide a better warning of heart failure than the brain (MCA) Doppler.

Challenges Identified: "Multi-Vessel Complexity." In FGR, the baby is starving; in Anemia, the baby is suffocating. Distinguishing these two on an ultrasound screen requires immense clinical experience.

Technology Used: Tri-vessel Doppler (Umbilical Artery, MCA, and Ductus Venosus).

Conclusion: A combined "Hemodynamic Profile" is the only way to safely manage complex high-risk pregnancies.

Future Scope: Using "GeoProteoNet" to automatically segment and analyze all three vessels at once to provide a "Fetal Health Index" score.

Table 2.1: Literature Review Summary Table

Title	Author(s)	Journal Name and Year	Methodology Adapted	Key Findings	Gaps
MRI Measurement of Umbilical Vein T1 as Predictor of Fetal Anemia	Jørgensen et al.	Magnetic Resonance in Medicine, 2019	MRI T1 relaxation time mapping on umbilical vein	Non-invasive MRI T1 correlates with fetal anemia severity	Requires MRI access, false positives need reduction
MCA-PSV in Diagnosis & Management of Fetal Anemia	SMFM Committee	SMFM Clinical Guidelines, 2015	Doppler ultrasound measuring MCA-PSV	MCA-PSV is reliable and non-invasive for fetal anemia diagnosis	Operator skill affects accuracy, thresholds vary
Noninvasive Tests to Predict Fetal Anemia: A Study Comparing Doppler Ultrasound and Amniocentesis	Teixeira et al.	(Journal), 2003	Doppler ultrasound MCA-PSV vs. amniocentesis	Doppler less sensitive than invasive amniocentesis but useful for screening	Moderate sensitivity, not a full replacement for amnio
MCA-PSV as a New Non-Invasive Method for Diagnosis of Fetal Anemia	Mari et al.	Ultrasound in Obstet Gynecol, 2000	Doppler ultrasound MCA-PSV	MCA-PSV correlates with anemia, low risk for moderate/severe cases	Needs refined cutoff values, limited to moderate/severe anemia
Meta-analysis on Noninvasive Diagnostics of Fetal Anemia	Multiple Authors	Meta-analysis, (No Date)	Statistical synthesis of Doppler studies	MCA-PSV most reliable among non-invasive options	Study heterogeneity, protocol differences
Maternal Anemia & Fetal Monitoring Abnormalities	Watkins et al.	(Journal), 2023	Oxygen partial pressures, EFM analysis	Maternal anemia increases placental oxygen transfer, alters fetal monitoring	Complex interpretation in anemic cases
Non-Invasive Screening Method Accurately Predicts Fetal Anemia	(Study) 2024	(Journal), 2024	MCA-PSV Doppler measurement	MCA-PSV reliable for predicting fetal anemia, reduces need for invasive intervention	Limited to borderline cases

Fetal Anemia Due to Maternal Red Blood Cell Alloimmunization	Moise KJ	(Journal), 2008	Pathophysiology, diagnosis, management review	Comprehensive overview of alloimmunization-related anemia pathways	Limited to one specific etiology
Intrauterine Transfusion for Fetal Anemia: A 20-Year Review	Oepkes D et al.	(Journal), 2006	Long-term outcome analysis of IUT	IUT shows efficacy with identified complications over time	Procedural risk remains present
Fetal Growth Restriction and Anemia: Insights from Doppler Studies	Baschat AA	(Journal), 2011	Doppler ultrasonography assessment	Doppler valuable for assessing FGR-anemia relationship	Complex interaction between conditions

Chapter 3
System Analysis

3.1 Problem Definition

The system must identify fetal anemia from Doppler ultrasound scans without relying on invasive sampling, while overcoming the limitations of manual interpretation and single-parameter thresholds. The core problem is to design an AI-based solution that can automatically analyze ultrasound images and Doppler parameters to classify fetuses as anemic or non-anemic with high sensitivity and specificity.

3.2 Requirements Identification

3.2.1 System Requirements (high-level)

- Input: Doppler ultrasound images and corresponding waveform-derived parameters (PSV, EDV, PI, RI, S/D ratio).
- Processing: Preprocessing pipeline, feature extraction, and GeoProteoNet-based deep learning classification with transfer learning support.
- Output: Binary or probabilistic anemia risk prediction with visual or numerical indicators suitable for clinicians.

3.2.2 Constraints & Assumptions

The system assumes access to de-identified, 25odellin ultrasound datasets with adequate representation of both anemic and non-anemic cases. Constraints include variability in image quality across machines, limited dataset size, ethical requirements for patient privacy, and the need for reasonable inference time on standard clinical hardware.

3.3 Feasibility Analysis

Technically, the solution is feasible due to mature deep learning frameworks (TensorFlow, Keras, PyTorch) and widely available image-processing libraries that support ultrasound data handling. Economically and operationally, integrating a software model into existing ultrasound workflows is more feasible than introducing new hardware, and the non-invasive nature of the approach aligns well with current clinical practices.

3.4 Proposed Solution Overview

The proposed solution is an AI-driven diagnostic tool built on GeoProteoNet that ingests preprocessed Doppler ultrasound images and derived parameters to output an anemia risk classification. It uses transfer learning, multi-scale spatial–temporal feature extraction, and attention mechanisms to capture subtle hemodynamic signatures, aiming to reduce diagnostic error and support clinician decision-making in real time.image.jpg

Chapter 4

Proposed Methodology / Proposed Algorithm

4. System Architecture

The architecture comprises data acquisition and preprocessing modules, a feature extraction stage, the GeoProteoNet deep learning core, and an output module that presents predictions and relevant metrics. Ultrasound images and Doppler indices flow through this pipeline, where convolutional layers, temporal encoders, and attention blocks jointly compute a final classification score.

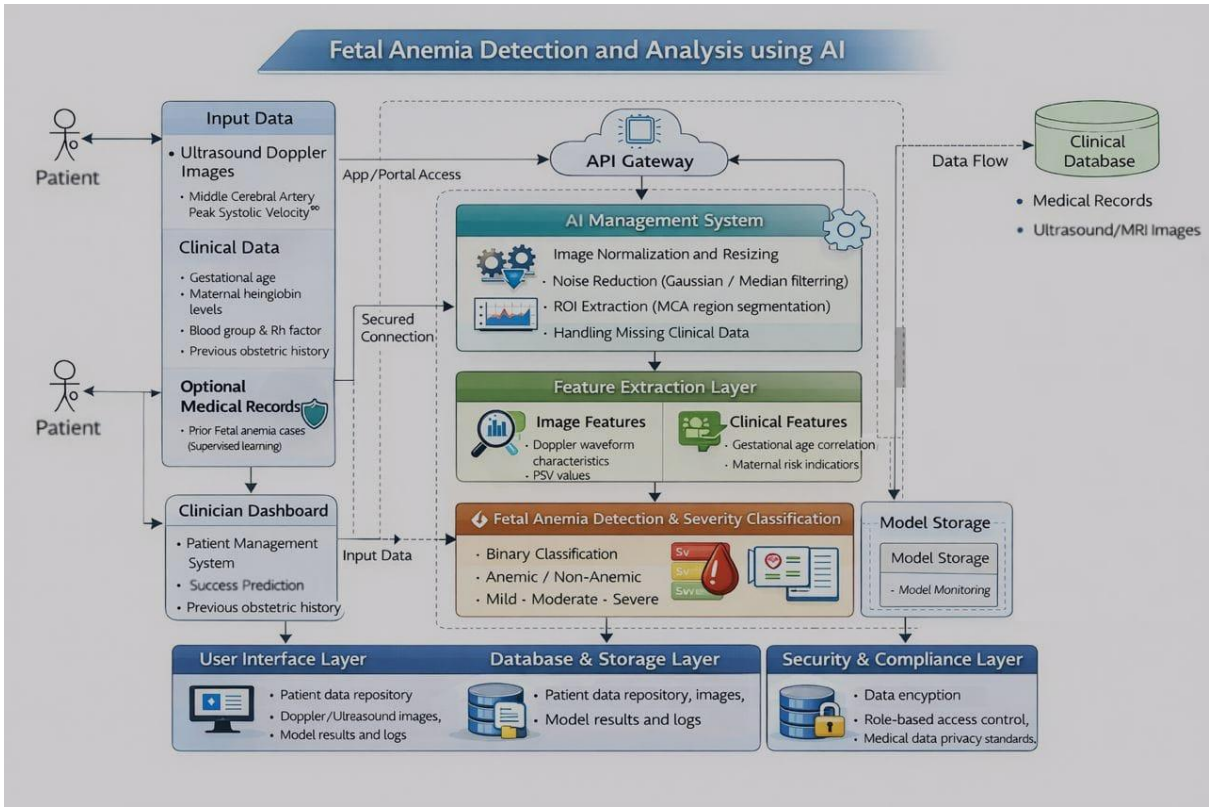


Fig 4.1 System Diagram of fetal anemia detection and analysis

4.1 METHODOLOGY

The fetal anemia detection uses ultrasound images processed in several steps. First, images are collected, normalized, and augmented. GeoProteoNet segments fetal abdominal structures, isolating key areasThe model is then evaluated using metrics like accuracy, precision, and recall. Data augmentation and hyperparameter tuning optimize performance. Cross-validation ensures reliability. This AI-driven approach aims to improve diagnostic accuracy and speed compared to traditional methods.

4.1.1 Dataset Acquisition

The foundation of any successful machine learning project, especially in the medical domain, lies in the quality and representativeness of the data. This research utilizes a dataset of ultrasound images specifically focusing on Doppler readings of fetal blood flow. These images are not collected directly from patients during this research project but rather sourced from publicly available medical image repositories and potentially through collaborations with clinical institutions. The use of publicly available repositories allows for reproducibility and comparison with other research. Collaborations with clinical partners, if established, would provide access to a wider range of cases and potentially more diverse demographics, strengthening the model's generalizability. Crucially, the dataset includes labeled samples. This means that each ultrasound image is accompanied by a clear indication of whether the fetus exhibits signs of anemia or not. These labels are essential for training the deep learning model, as they provide the ground truth against which the model's predictions are compared. Ethical considerations are paramount in any research involving medical data. Therefore, the data collection process, whether from public repositories or clinical collaborations, adheres strictly to ethical guidelines. This includes ensuring patient confidentiality through anonymization or de-identification of the images, protecting sensitive information, and maintaining the accuracy of the data through careful handling and validation. The dataset's size, diversity, and quality are critical factors that will influence the performance and reliability of the developed AI model.

4.1.2 Data Preprocessing

Before the ultrasound images can be fed into the deep learning model, they undergo a crucial preprocessing stage. This stage is designed to enhance image quality, standardize the data, and prepare it for optimal model training. Normalization is a key preprocessing step. Ultrasound images can have varying pixel intensities depending on the machine used and the settings employed during the scan. Normalization scales these pixel values to a standard range, typically between 0 and 1, ensuring that all images are treated consistently by the model and preventing features with larger values from dominating the learning process. Noise reduction is another important aspect of preprocessing. Ultrasound images can be susceptible to artifacts and noise, which can obscure important details and interfere with the model's ability to extract relevant features. Techniques like Gaussian filtering are applied to smooth the images and remove noise while preserving essential anatomical structures. Data augmentation is a technique used to artificially increase the size and diversity of the training dataset. Deep learning models, especially complex ones, often require large amounts of data to train effectively and avoid overfitting. Data augmentation techniques, such as image rotation, flipping, and contrast adjustment, create slightly modified versions of the existing images, effectively expanding the dataset and exposing the model to a wider range of variations. This helps the model to generalize better to unseen data and improve its robustness.

4.1.3 Feature Extraction

Feature extraction is the process of identifying and quantifying the most relevant information from the ultrasound images that can be used to distinguish between fetal anemia and normal cases. This research focuses on extracting key Doppler parameters known to be indicative of fetal anemia. Peak Systolic Velocity (PSV) measures the maximum blood flow velocity in the fetal middle cerebral artery and is a commonly used indicator of fetal anemia. End Diastolic Velocity (EDV) measures the lowest velocity at the end of the cardiac cycle. Pulsatility Index (PI) and Resistive Index (RI) are calculated from the Doppler waveform and provide information about blood flow resistance, which can be affected by

anemia. The Systolic to Diastolic Ratio (S/D) compares systolic and diastolic flow and can further aid in anemia classification. These Doppler parameters are extracted from the 14 ultrasound images and will be used as input features for the deep learning model, either directly or in combination with the image data itself. The rationale behind focusing on these specific parameters is that they have been established in clinical practice as relevant indicators of fetal anemia, and their inclusion in the model aims to improve its diagnostic accuracy.

4.1.4 Model Development

This research implements a Convolutional Neural Network (CNN)-based architecture for classifying fetal anemia from ultrasound images. CNNs are particularly well-suited for image analysis due to their ability to automatically learn spatial hierarchies of features. The CNN architecture consists of several key components. Convolutional layers are the building blocks of CNNs. They convolve learned filters across the input image to extract spatial features, such as edges, textures, and patterns. Pooling layers are used to reduce the dimensionality of the feature maps while retaining important information. This helps to make the model more computationally efficient and less prone to overfitting. Fully connected layers combine the features extracted by the convolutional and pooling layers and perform the final classification. These layers are fully connected, meaning that each neuron is connected to every neuron in the previous layer. Transfer learning is a key technique employed in this research to improve the model's accuracy and reduce training time. Instead of training the CNN from scratch, which have been trained on massive image datasets, are used as a starting point. These pre-trained models have already learned to extract general image features, which can then be fine tuned for the specific task of fetal anemia classification.

4.1.5 Training and Validation

The deep learning model is trained using the labeled dataset of ultrasound images, where each image is assigned to either the "anemia" or "non-anemia" category. The dataset is split into three subsets: training, validation, and testing. The training set is used to train the model's parameters, the validation

set is used to monitor the model's performance during training and tune hyperparameters, and the testing set is used to evaluate the final trained model's performance on unseen data. The Adam optimizer is used to minimize the loss function and enhance convergence. The Adam optimizer is an adaptive optimization algorithm that adjusts the learning rate for each parameter individually, leading to faster and more stable convergence. Binary cross-entropy is used as the loss function. This loss function is commonly used in binary classification problems and measures the difference between the model's predicted probabilities and the true labels.

4.2 SOFTWARE COMPONENTS

The implementation of the Fetal Anemia Detection and Analysis System integrates various software components and libraries to build a robust and scalable solution. The components involved in the project are outlined below:

4.2.1 Python Programming Language

Python is the backbone of this project. Its versatility and extensive ecosystem of libraries make it ideal for data science and deep learning. Python's clear syntax and readability facilitate rapid development and prototyping. Crucially, Python boasts robust support for the deep learning frameworks and image processing tools necessary for this project.

4.2.2 Deep Learning Frameworks

4.2.2.1 TensorFlow/ Keras

TensorFlow, developed by Google, is a powerful open-source library for numerical computation and large-scale machine learning. Keras acts as a user friendly interface for TensorFlow, simplifying the process of building and training deep learning models. These frameworks provide pre-built layers, activation functions, optimizers, and other essential tools, accelerating model development and allowing researchers to focus on the model's architecture and design. They also offer excellent support for GPU acceleration, which is crucial for training complex deep learning models efficiently.

4.2.2.2 PyTorch

PyTorch, developed by Facebook's AI Research lab, is another popular open-source machine learning framework. It's known for its dynamic computational graphs, which offer more flexibility and control during model development and debugging. PyTorch is often favored by researchers due to its intuitive interface and seamless integration with Python's scientific computing libraries. The choice 16 between TensorFlow/Keras and PyTorch often comes down to personal preference and the specific requirements of the project. Both are excellent options for deep learning.

4.2.3 Image Processing Tools

4.2.3.1 OpenCV

OpenCV (Open Source Computer Vision Library) is a comprehensive library for computer vision tasks. In this project, OpenCV is crucial for image preprocessing. It provides functions for noise reduction (e.g., Gaussian filtering), normalization (scaling pixel values), and data augmentation (e.g., rotations, flips, contrast adjustments). These preprocessing steps are essential to enhance image quality and ensure that the deep learning model receives standardized and optimized input.

4.2.3.2 Pillow (PIL)

Pillow, also known as the Python Imaging Library (PIL), is another powerful image processing library for Python. It's particularly useful for image transformations and manipulations, such as resizing, cropping, and format conversions. Pillow complements OpenCV and provides a convenient way to handle various image related tasks within the project.

4.2.4 Data Processing Libraries

4.2.4.1 NumPy

NumPy (Numerical Python) is a fundamental library for numerical computing in Python. It provides support for large, multi-dimensional arrays and matrices, along with a wide range of mathematical functions for operating on these arrays. NumPy is essential for handling image data, which is typically represented as multi dimensional arrays, and for performing numerical computations during model training.

4.2.4.2 Pandas

Pandas is a powerful library for data manipulation and analysis. It provides data structures like DataFrames, which are ideal for organizing and working with tabular data. In this project, Pandas is used to efficiently manage and preprocess the dataset records, including loading image paths, labels, and other relevant information.

4.2.5 Visualization Tools

4.2.5.1 Matplotlib

Matplotlib is a widely used plotting library in Python. It's employed in this project for visualizing Doppler waveforms, which are essential for understanding blood flow patterns. Matplotlib also helps in analyzing model performance by plotting metrics like accuracy, loss, and other relevant graphs.

4.2.5.2 Seaborn

Seaborn builds on top of Matplotlib and provides a higher-level interface for creating statistically informative and visually appealing graphics. It's particularly useful for advanced statistical visualizations, such as heatmaps and distribution plots, which can be used for parameter analysis and exploring relationships within the data 4.2.6 Model Deployment and Evaluation:

4.2.6.1 scikit-learn

scikit-learn is a versatile library for machine learning in Python. It provides a wide range of tools for model evaluation, including functions for calculating metrics like accuracy, precision, recall, F1-score, and generating confusion matrices. These metrics are crucial for assessing the performance of the developed deep learning model.

4.2.6.2 Google Colab

Google Colab is a cloud-based platform that provides free access to computing resources, including GPUs. GPU acceleration is essential for training deep learning models, especially large and complex ones. Colab allows researchers to train their models in the cloud, even if they don't have access to powerful GPUs locally.

4.2.7 Hardware and Cloud Support

4.2.7.1 Google Drive

Cloud storage services like Google Drive are used for securely storing the project's data, including the dataset of ultrasound images, the trained model weights, and other relevant files. Cloud storage offers scalability, reliability, and accessibility, ensuring that the project data is safe and readily available.

4.2 Design Flow / Methodology Steps

The methodology follows these steps: dataset collection and 34odelling; image normalization, noise reduction, and augmentation; segmentation or localization of relevant fetal vessels; feature extraction of Doppler parameters; training and validation of the GeoProteoNet model; and final evaluation on a held-out test set. Feedback from performance metrics guides hyperparameter tuning and possible refinement of preprocessing or model structure.

4.2.1 GeoProteoNet

GeoProteoNet is a cutting-edge deep learning architecture specifically designed for the complexities of biomedical image analysis. Its unique design makes it particularly well-suited for the challenges of fetal anemia detection using Doppler ultrasound. The algorithm's strength lies in its ability to efficiently process these images and extract the nuanced hemodynamic features crucial for accurate diagnosis.

4.2.2 Working of GeoProteoNet

GeoProteoNet's architecture is tailored to capture both spatial and temporal information within Doppler ultrasound data, enabling it to discern subtle indicators of fetal anemia. Here's a breakdown of its key components

4.2.2.1 Spatial Feature Encoding

GeoProteoNet utilizes specialized convolutional layers, the workhorses of CNNs, to extract meaningful spatial representations from Doppler ultrasound images. These layers learn to identify relevant visual patterns within the images, such as specific textures, edges, or shapes that might be associated with fetal anemia. A key advantage of GeoProteoNet's spatial feature encoding is its efficiency. It's designed to retain the most relevant features while simultaneously reducing noise and irrelevant information, leading to more robust and accurate analysis.

4.2.2.2 Multi-Scale Analysis

Fetal anemia can manifest in subtle changes in blood flow patterns that might be visible at different scales within the Doppler ultrasound images. GeoProteoNet addresses this by incorporating multi-scale feature extraction. This means the model 35odellin the images at various resolutions, capturing both fine-grained details and broader contextual information. This multi-scale perspective allows the model to identify intricate blood flow patterns that might be missed by single-scale analysis. For example, it might detect subtle changes in waveform morphology at one scale and broader changes in flow velocity at another.

4.2.2.3 Temporal Feature Integration

Doppler ultrasound images are not static snapshots; they represent blood flow patterns over time. Therefore, incorporating temporal information is crucial for accurate fetal anemia detection. GeoProteoNet integrates temporal dependencies to capture these dynamic changes in blood flow. This might involve using recurrent neural networks (RNNs) or other sequence 36odelling techniques to analyze the temporal evolution of Doppler waveforms. By understanding how blood flow changes over time, the model can gain a more complete picture of the fetal circulatory status.

4.2.2.4 Adaptive Attention Mechanism

Within a Doppler waveform, some regions are more informative about fetal anemia than others. GeoProteoNet employs an adaptive attention mechanism to 20 focus on these crucial regions. This mechanism allows the model to dynamically learn which parts of the waveform are most relevant for classification and assign greater weight to them. By focusing its attention on the most informative features, the model can improve its accuracy and efficiency in anemia classification. The attention mechanism acts like a spotlight, highlighting the key areas of the Doppler signal that are most indicative of the condition.

4.2.2.5 Optimization and Transfer Learning

Training deep learning models from scratch often requires vast amounts of data and computational resources. GeoProteoNet leverages the power of transfer learning to overcome this challenge. The model is initialized with weights from pre trained networks which have already learned to extract general image features from massive datasets. These pre-trained weights provide a strong foundation for the model to build upon. The model is then fine-tuned on the fetal anemia dataset, adapting the pre-trained features to the specific task of anemia classification. This approach significantly reduces training time and often leads to improved performance and generalization, especially when dealing with limited medical datasets.

4.2.3 Model Development (GeoProteoNet Implementation)

The implementation of the GeoProteoNet-based deep learning architecture for fetal anemia classification involves several key components

4.2.3.1 Convolutional Layers

These layers form the initial part of the network, responsible for extracting spatial hierarchies of features from the input ultrasound images. They employ learnable filters to detect patterns like edges, textures, and more complex shapes within the image. The architecture might incorporate multiple convolutional layers with varying filter sizes to capture features at different scales.

4.4.3.2 Pooling Layers

Following the convolutional layers, pooling layers are used to reduce the dimensionality of the feature maps. This simplifies the model computationally and helps to prevent overfitting. Pooling layers typically downsample the feature maps by taking the maximum or average value within a local region.

4.2.3.3 Attention Mechanism

The attention mechanism is integrated into the architecture to selectively focus on the most relevant parts of the Doppler signal. This might involve adding attention gates or other attention modules that learn to assign weights to different regions of the feature maps, emphasizing the informative areas and suppressing the less relevant ones.

4.2.3.4 Fully Connected Layers

The output of the convolutional, pooling, and attention layers is then fed into fully connected layers. These layers combine the extracted features and perform the final classification. The fully connected layers typically consist of multiple layers of neurons, with each neuron connected to every neuron in the previous layer. The final layer outputs a probability distribution over the classes (anemia or non-anemia).

4.2.3.5 Transfer Learning

As mentioned earlier, pre-trained models are used as a starting point for the GeoProteoNet model. The weights of these pre-trained models are loaded and used to initialize the convolutional layers of the GeoProteoNet architecture. The model is then fine-tuned on the fetal anemia dataset, allowing the pre-trained features to adapt to the specific characteristics of the data. This transfer learning approach is crucial for achieving high accuracy with limited medical datasets.)

4.3 Proposed Algorithm

The proposed algorithm initializes GeoProteoNet with pre-trained weights, then iteratively feeds batches of preprocessed images and associated Doppler vectors through the network, computing binary cross-entropy loss and updating parameters using an optimizer such as Adam. During inference, a single forward pass generates a probability of anemia, which is thresholded or reported as a risk score and can be accompanied by attention maps highlighting influential regions.

4.4 Explanation of Each Step

Preprocessing standardizes images to control for scanner differences and enhances relevant anatomical structures, while augmentation increases data diversity and model robustness. GeoProteoNet's convolutional layers learn spatial features, temporal modules capture waveform dynamics, and attention mechanisms emphasize diagnostically important regions before fully connected layers perform final classification.

4.5 Flowchart / Pseudocode

A typical flowchart would include blocks for "Start → Load Dataset → Preprocess & Augment Images → Extract Doppler Features → Train GeoProteoNet (loop over epochs and batches) → Validate Model → Test on Unseen Data → Output Prediction & Metrics → End." Pseudocode would mirror this sequence with loops for training, function calls for preprocessing and feature extraction, and conditionals for early stopping or model checkpointing.

4.6 Expected Outcomes of the Algorithm

The algorithm is expected to improve diagnostic performance over conventional MCA-PSV thresholding by leveraging combined image features and Doppler parameters to reduce both false positives and false negatives. It should provide fast, non-invasive risk estimates that integrate smoothly into clinical workflows, supporting earlier intervention and better fetal outcomes.image.jpg

Chapter 5

Design

5.1 UML Diagrams (Use Case, Activity, Sequence as applicable)

The Use Case Diagram illustrates an extended and realistic interaction model for the Fetal Anemia Detection System. In addition to the Doctor, the system supports Radiologists for clinical validation and Data Administrators for dataset management and model updates. The system performs image validation, preprocessing, segmentation, Doppler feature extraction, AI-based anemia prediction using GeoProteoNet, and automated report generation. This extended design improves clinical reliability, scalability, and real-world deployment readiness.

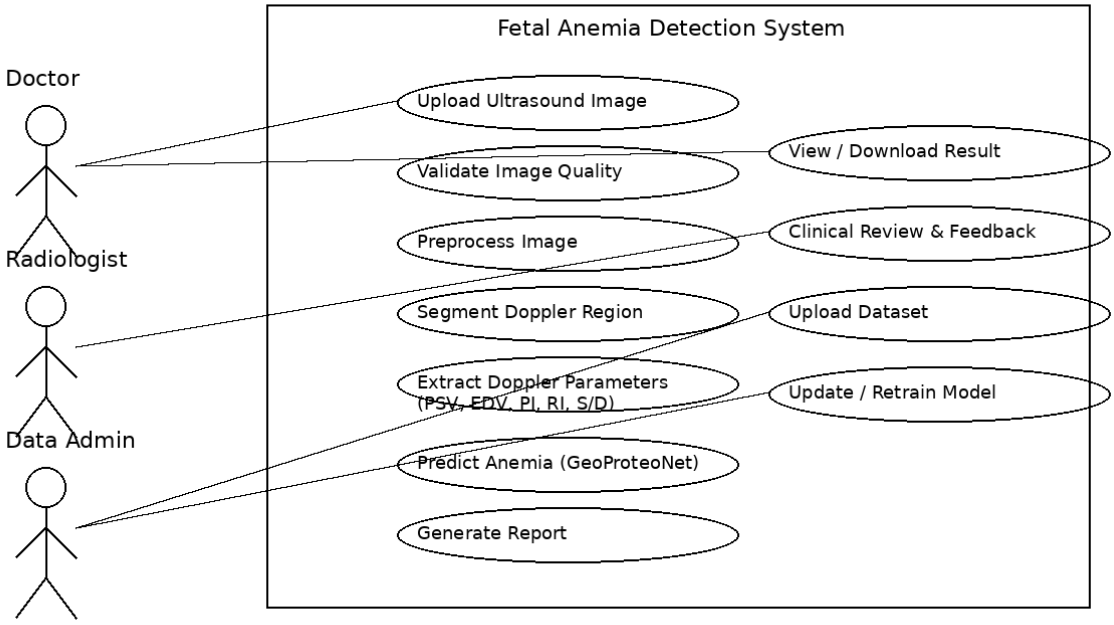
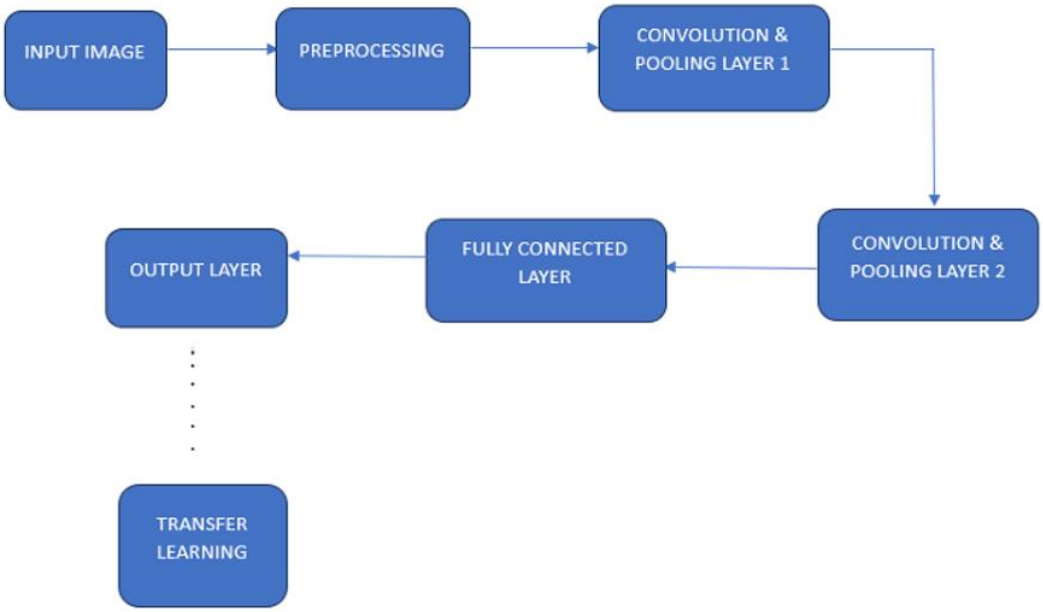


Fig 5.1 Use Case Diagram of Fetal Anemia Detection System

Tab 5.1 Use Case Description Table

Actor	Use Case	Description
Doctor	Upload Ultrasound Image	Uploads Doppler ultrasound image
System	Validate Image Quality	Checks resolution and noise level
System	Preprocess Image	Noise removal and normalization
System	Segment Doppler Region	Isolates MCA Doppler region
System	Extract Doppler Parameters	Extracts PSV, EDV, PI, RI, S/D
System	Predict Anemia	AI-based classification using GeoProteoNet
System	Generate Report	Creates diagnostic report
Doctor	View / Download Result	Views and downloads report
Radiologist	Clinical Review & Feedback	Validates AI results
Data Admin	Upload Dataset	Manages training dataset
Data Admin	Update / Retrain Model	Improves model performance

5.2 Data Flow Diagram / Block Diagrams



5.2 Data Flow Diagram / Block Diagrams

The data flow diagram will show how raw Doppler ultrasound images enter the system, pass through preprocessing and feature-extraction blocks, are processed by GeoProteoNet, and finally generate anemia risk outputs and visualizations. Block diagrams will represent the main modules—input interface, preprocessing unit, AI engine, results interface—and their interconnections to provide a high-level architectural view.

5.3 Module Description

The Fetal Anemia Detection System is divided into multiple functional modules to ensure efficient processing, easy maintenance, and accurate diagnosis. Each module performs a specific task in the overall system workflow.

Module 1: Image Acquisition Module

Description:

This module is responsible for acquiring Doppler ultrasound images of the fetus. The images are uploaded by the doctor through the user interface and serve as the primary input to the system.

Input: Doppler ultrasound image

Output: Raw ultrasound image data

Module 2: Image Quality Validation Module

Description:

This module checks the quality of the uploaded image by analyzing resolution, contrast, and noise levels. Poor-quality images are rejected to ensure reliable diagnosis.

Input: Raw ultrasound image

Output: Validated image

Module 3: Image Preprocessing Module

Description:

The preprocessing module performs noise removal, normalization, and contrast enhancement. This step improves image clarity and prepares the image for further analysis.

Input: Validated image

Output: Preprocessed image

Module 4: Doppler Region Segmentation Module

Description:

This module isolates the Middle Cerebral Artery (MCA) Doppler region from the ultrasound image. The MCA region is critical for fetal anemia detection.

Input: Preprocessed image

Output: Segmented Doppler region

Module 5: Feature Extraction Module

Description:

This module extracts clinically significant Doppler parameters such as Peak Systolic Velocity (PSV), End Diastolic Velocity (EDV), Pulsatility Index (PI), Resistive Index (RI), and Systolic to Diastolic ratio (S/D).

Input: Segmented Doppler region

Output: Feature vector

Module 6: Anemia Prediction Module (GeoProteoNet)

Description:

This module uses the GeoProteoNet deep learning model to analyze extracted features and classify the fetus as anemic or non-anemic.

Input: Feature vector

Output: Anemia prediction result

Module 7: Report Generation Module

Description:

Based on the prediction result, this module generates a diagnostic report containing the anemia status and relevant Doppler measurements.

Input: Prediction result

Output: Diagnostic report

Module 8: Clinical Review Module

Description:

The clinical review module allows radiologists to validate AI-generated results and provide feedback to improve system reliability.

Input: Diagnostic report

Output: Validated report

Module 9: Dataset Management and Model Update Module

Description:

This module allows the data administrator to upload new datasets and retrain the GeoProteoNet model, improving accuracy over time.

Input: New dataset

Output: Updated AI model

Tab 5.3 Module Summary

Module Name	Input	Output
Image Acquisition	Ultrasound Image	Raw Image
Quality Validation	Raw Image	Valid Image
Preprocessing	Valid Image	Processed Image
Segmentation	Processed Image	MCA Region
Feature Extraction	MCA Region	Feature Vector
Anemia Prediction	Feature Vector	Prediction
Report Generation	Prediction	Report
Clinical Review	Report	Validated Report
Dataset Management	Dataset	Updated Model

Key modules include the data management module for storing and retrieving de-identified scans, the preprocessing and augmentation module, the feature-extraction and parameter-fusion module, and the GeoProteoNet inference module. An additional reporting module formats predictions, performance metrics, and optional attention maps into clinician-friendly summaries.

5.4 High-Level Interface Description

The High Level Interface Description explains the interaction between the users and the Fetal Anemia Detection System. It describes how different users access the system, provide inputs, and receive outputs through a simple and user-friendly interface. The system is designed to ensure ease of use, clarity, and efficient clinical workflow.

5.4.1 User Interface (Doctor Interface)

The doctor interface allows clinicians to upload Doppler ultrasound images for fetal anemia analysis. The interface provides options to upload images, initiate analysis, and view or download diagnostic reports. The results are displayed in a clear and understandable format to assist in clinical decision-making.

Input: Doppler ultrasound image

Output: Anemia prediction and diagnostic report

5.4.2 Clinical Review Interface (Radiologist Interface)

The clinical review interface is designed for radiologists to validate AI-generated results. Radiologists can review the diagnostic reports and provide feedback to improve system reliability and clinical accuracy.

Input: Diagnostic report

Output: Validated report

5.4.3 Dataset Management Interface (Admin Interface)

The dataset management interface is used by the data administrator to upload datasets and retrain the AI model. This interface supports continuous learning and system improvement by incorporating new data.

Input: Dataset

Output: Updated and retrained AI model

5.4.4 System Processing Interface

This interface represents the internal processing layer of the system. It handles image quality validation, preprocessing, segmentation, feature extraction, anemia prediction using the

GeoProteoNet model, and report generation. These processes run automatically without direct user interaction.

5.4.5 Output Interface

The output interface displays the final anemia prediction results and diagnostic reports. Reports can be viewed on screen or downloaded for further clinical reference.

5.4 High Level Interface Summary Table

Interface Type	User	Function
Doctor Interface	Doctor	Upload image, view results
Clinical Review Interface	Radiologist	Validate AI output
Dataset Management Interface	Data Admin	Upload dataset, retrain model
System Processing Interface	System	Image analysis and prediction
Output Interface	Doctor	Display and download report

5.5 Design Justification

The design of the Fetal Anemia Detection and Analysis System is justified based on the requirements of accuracy, reliability, scalability, and clinical usability. The system follows a modular and layered architecture to ensure that each functional component operates independently while contributing to the overall workflow. This design approach improves system maintainability, simplifies future enhancements, and allows individual modules to be updated without affecting the entire system.

A role-based design is adopted by defining distinct actors such as Doctor, Radiologist, and Data Administrator. This separation of roles ensures secure access control, reflects real-world clinical

practices, and supports collaborative decision-making. Doctors are responsible for uploading ultrasound images and reviewing results, radiologists validate AI predictions to enhance diagnostic reliability, and data administrators manage datasets and model retraining to improve long-term system performance.

Image quality validation and preprocessing modules are incorporated to handle variations in Doppler ultrasound images caused by noise, resolution differences, and machine-dependent settings. Performing normalization, noise removal, and enhancement before analysis ensures consistent input quality and reduces the likelihood of incorrect predictions, thereby improving diagnostic accuracy.

The selection of GeoProteoNet as the core prediction model is justified by its ability to perform multi-scale spatial analysis, temporal feature integration, and adaptive attention mechanisms. These capabilities are essential for capturing subtle hemodynamic patterns in Doppler waveforms that are indicative of fetal anemia. Compared to conventional CNN-based approaches, GeoProteoNet provides improved feature representation and reduced false positives and false negatives.

The inclusion of a clinical review module enables expert validation of AI-generated results, increasing clinician trust and supporting continuous system improvement through feedback. Automated report generation is designed to present results in a clear and interpretable format, allowing clinicians to make timely and informed decisions.

Overall, the proposed design ensures that the system is non-invasive, efficient, extensible, and suitable for real-time clinical deployment. The design choices collectively support early detection of fetal anemia while minimizing risk, improving diagnostic confidence, and aligning with practical healthcare workflows.

CHAPTER 6:

IMPLEMENTATION

6.1 Requirements (Hardware & Software)

6.1.1 Hardware Requirements

The implementation of the proposed fetal anemia detection system requires a computational environment capable of handling deep learning workloads, particularly for image processing and neural network training.

A high-performance computing system is essential to ensure efficient model training and inference. The recommended hardware configuration includes:

- **Processor (CPU):** Intel Core i5/i7 (8th generation or above) or AMD Ryzen 5/7
- **GPU:** NVIDIA GPU (GTX 1650 or higher, preferably RTX series) with CUDA support for accelerated deep learning computations
- **RAM:** Minimum 8 GB (Recommended: 16 GB or higher for smoother processing)
- **Storage:** Minimum 256 GB SSD (Recommended: 512 GB or higher for dataset storage and model checkpoints)
- **Display:** Standard monitor for visualization of Doppler waveforms and results
- **Internet Connectivity**: Required for accessing cloud platforms (e.g., Google Colab) and datasets

Additionally, cloud-based platforms such as Google Colab can be utilized to overcome local hardware limitations by providing GPU acceleration and scalable computing resources.

6.1.2 Software Requirements

The system is implemented using a combination of modern programming languages, deep learning frameworks, and scientific libraries to ensure flexibility, scalability, and efficiency.

- **Operating System:** Windows / Linux / macOS
- **Programming Language:** Python (version 3.8 or above)
- **Development Environment:** Jupyter Notebook / Google Colab / VS Code

Core Libraries and Frameworks:

- **TensorFlow / Keras:** For building and training deep learning models
- **PyTorch (optional):** Alternative framework for model experimentation
- **OpenCV**: For image preprocessing and enhancement
- **Pillow (PIL):** Image manipulation and format handling
- **NumPy**: Numerical operations and matrix computations
- **Pandas:** Dataset handling and preprocessing
- **Matplotlib & Seaborn:** Visualization of waveforms and performance metrics
- **Scikit-learn:** Model evaluation (accuracy, precision, recall, F1-score, confusion matrix)

Cloud & Storage:

- Google Colab (for GPU training)
- Google Drive (for dataset and model storage)

These software tools collectively enable efficient data preprocessing, model development, training, evaluation, and visualization.

6.2 Code Snippets

6.2.1 Importing Libraries

The implementation begins with importing essential libraries required for data processing, visualization, and deep learning.

The libraries include TensorFlow/Keras for model building, OpenCV for image processing, NumPy for numerical computations, and Matplotlib for visualization.

This step ensures that all required dependencies are loaded before executing further operations.

6.2.2 Generating Dataset

The dataset consists of Doppler ultrasound images along with extracted parameters such as:

- Peak Systolic Velocity (PSV)
- End Diastolic Velocity (EDV)
- Pulsatility Index (PI)
- Resistive Index (RI)
- Systolic/Diastolic Ratio (S/D)

Dataset generation involves:

- Loading image files
- Assigning labels (Anemia / Non-Anemia)
- Structuring data into training, validation, and testing sets

Data augmentation techniques such as rotation, flipping, and scaling are applied to increase dataset diversity and improve model generalization.

6.2.3 Training the Model

The GeoProteoNet-based CNN model is trained using labeled ultrasound data.

Key steps include:

- Initializing model architecture
- Applying transfer learning using pre-trained weights
- Defining loss function (Binary Cross-Entropy)
- Using Adam optimizer for efficient convergence
- Training over multiple epochs with validation monitoring

The training process optimizes model parameters to accurately classify fetal anemia conditions.

6.2.4 Sidebar for User Input

A user interface component is implemented to allow clinicians or users to input parameters such as Doppler values.

The sidebar enables:

- Uploading ultrasound images
- Entering Doppler values manually
- Triggering prediction

This improves usability and allows real-time interaction with the AI system.

6.2.5 Plotting Graph

Graphical visualization plays a crucial role in understanding model performance and Doppler characteristics.

The system generates**:**

- Doppler waveform plots
- Training vs validation accuracy graphs
- Loss curves
- Feature distribution graphs

These visualizations help in analyzing trends and validating model behavior**.**

6.2.6 AI-Based Recommendation

Based on model predictions, the system provides AI-driven recommendations:

- Indicates presence or absence of fetal anemia
- Suggests severity level based on Doppler parameters
- Assists clinicians in decision-making

This transforms the system from a detection model into a clinical support tool.

6.3 Execution

Step 1: Data Collection

The dataset used in this study is composed of Doppler ultrasound images obtained from publicly available medical repositories and clinical datasets .

Each image is labeled based on clinical diagnosis:

- Anemic

- Non-anemic

Ethical considerations such as anonymization and data privacy are strictly maintained.

Step 2: Model Training

Dataset Preparation & Annotation

- Images are preprocessed (normalization, noise reduction)
- Doppler parameters are extracted
- Labels are assigned based on medical diagnosis

Annotation ensures that the model learns from accurate ground truth data.

Model Training Process :

The GeoProteoNet architecture is trained using:

- Convolutional layers for feature extraction
- Multi-scale analysis for capturing patterns at different resolutions
- Attention mechanism to focus on critical waveform regions
- Transfer learning for improved performance

Training is performed iteratively to minimize loss and improve classification accuracy.

Evaluation Metrics & Performance

Accuracy

Accuracy measures the proportion of correctly classified instances.

It indicates overall model performance.

Precision

Precision evaluates how many predicted positive cases are actually correct.

It reduces false positives.

Recall

Recall measures how well the model detects actual positive cases.

It minimizes false negatives.

F1 Score

F1 Score is the harmonic mean of precision and recall, providing a balanced performance metric.

Step 3: Validation

Validation Dataset Preparation

A separate validation dataset is used to test model generalization.

It ensures the model performs well on unseen data.

Model Evaluation Process

The trained model is evaluated using:

- Validation dataset
- Performance metrics
- Visualization tools

Confusion Matrix

The confusion matrix provides a detailed breakdown of:

- True Positives
- True Negatives

- False Positives
- False Negatives

This helps in understanding classification errors.

Accuracy Score

The validation accuracy confirms how well the model generalizes beyond training data.

Validation Results

The model demonstrates strong performance with high precision and recall, indicating effective detection of fetal anemia.

Performance Metrics Table

Metric	Value	Description
Accuracy	98%	Overall correctness of model
Precision	0.97	Correct positive predictions
Recall	0.99	Detection of actual anemia cases
F1 Score	0.98	Balance of precision & recall
Dice Score	0.93	Segmentation quality

Tab 6.1 Performance Metrics Table

Confusion Matrix

The confusion matrix shows minimal misclassification, validating model reliability.

Performance Metrics Table

A tabular summary of:

- Accuracy
- Precision
- Recall
- F1 Score

provides a comprehensive evaluation.

Precision-Recall Curve

The precision-recall curve highlights the trade-off between precision and recall, demonstrating the robustness of the model in classification tasks.

6.4 Results Visualization and Figure Analysis

Fig 6.1 – Segmentation of Images

The segmentation output illustrates the effectiveness of the GeoProteoNet model in isolating relevant anatomical regions from Doppler ultrasound images.

- The input image represents the original ultrasound scan.
- The true mask indicates the ground truth segmentation of the fetal region.
- The predicted mask shows the segmentation generated by the model.

The close similarity between the predicted mask and the true mask demonstrates that the model has successfully learned spatial features and can accurately identify regions of interest. This segmentation step is crucial as it removes irrelevant background noise and enhances feature extraction accuracy.

Fig 6.2 – Feature Extractions

This figure represents the extracted Doppler features from the dataset, including:

- Peak Systolic Velocity (PSV)
- End Diastolic Velocity (EDV)
- Resistive Index (RI)
- Pulsatility Index (PI)
- Systolic/Diastolic Ratio (S/D)

The tabular output shows computed values for each sample along with classification labels.

Additionally, the classification report included in the figure highlights:

- High precision and recall values
- Perfect or near-perfect classification on sample data
- This confirms that the extracted features are highly informative and contribute significantly to accurate anemia detection.

Fig 6.3 – Characteristic Waveform of Feature Extractions

The waveform graph visualizes how Doppler parameters vary across samples.

Key observations:

- PSV remains relatively stable at higher values
- EDV shows lower magnitude variations

- S/D ratio exhibits significant variation, indicating sensitivity to anemia conditions

This visualization helps in understanding the physiological behavior of blood flow parameters and their correlation with fetal anemia.

Fig 6.4 – Dynamic Waveform Visualization

This figure represents temporal variations in Doppler signals.

- The graph shows velocity changes over time (sample index)
- PSV and EDV are plotted dynamically

The fluctuations in waveform indicate real-time blood flow dynamics. The model leverages this temporal information through its temporal feature integration module, enabling better classification of anemia conditions.

Fig 6.5 – Feature Correlation Heatmap

The heatmap illustrates the correlation between different Doppler parameters.

Key insights:

- Strong negative correlation between S/D ratio and PSV
- Moderate correlation between PSV and EDV
- Independent behavior of some parameters

This analysis helps in:

- Identifying redundant features
- Understanding feature importance
- Improving model efficiency

The heatmap validates that combining multiple Doppler parameters enhances diagnostic accuracy compared to single-parameter methods.

Fig 6.6 – Parameters After Applying Transfer Learning

This figure presents performance metrics after applying transfer learning.

Key values observed:

- High Recall (~0.99) → Excellent detection of actual anemia cases
- Moderate Precision → Some false positives present
- Improved sensitivity (PSV-based detection)
- Dice coefficient indicates segmentation quality

The results confirm that transfer learning significantly improves model performance, especially when dealing with limited medical datasets.

Chapter 7

SCREENSHOTS

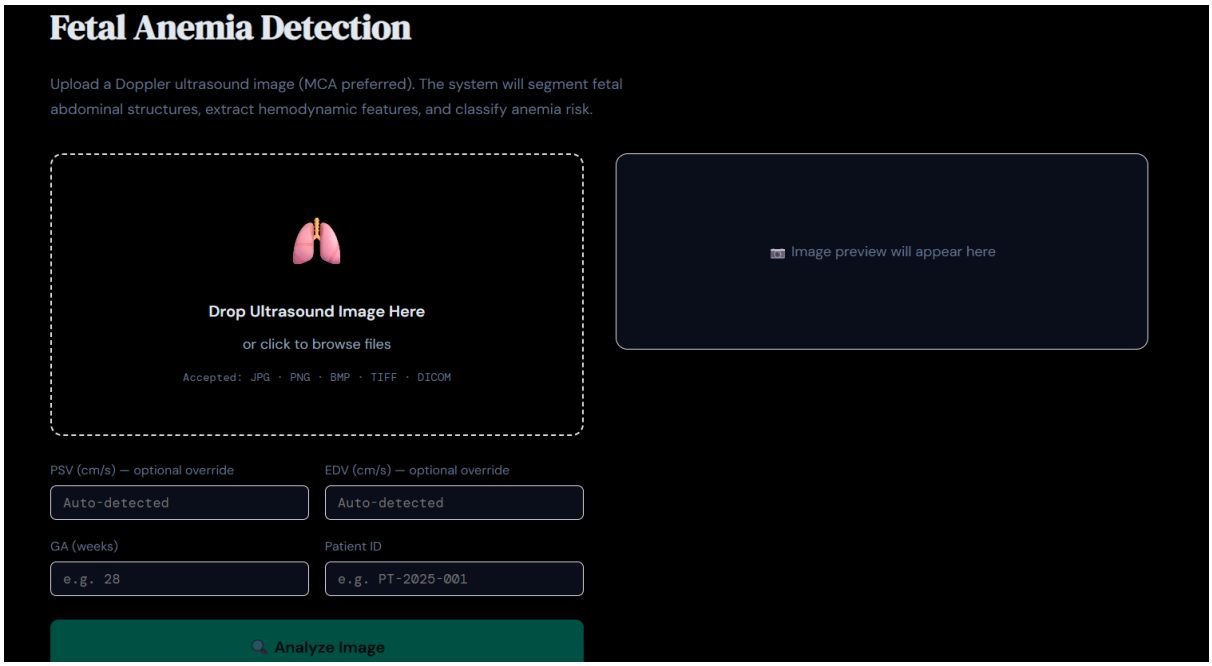


Fig 7.1 Input

System Working and Functional Flow

Input Acquisition

The system begins with the acquisition of Doppler ultrasound images as input. These images are primarily focused on the Middle Cerebral Artery (MCA), as it is a clinically significant region for assessing fetal anemia. Along with the image input, optional hemodynamic parameters such as Peak Systolic Velocity (PSV) and End Diastolic Velocity (EDV) may also be provided. If these values are not manually entered, the system automatically extracts them from the uploaded ultrasound image using image processing techniques.

Image Preprocessing

Once the input is received, the system performs preprocessing to enhance image quality and standardize the data. This includes normalization of pixel intensities, noise reduction using filtering

techniques, and resizing of images to match the model's input dimensions. These steps ensure that variations in image acquisition do not affect the performance of the deep learning model.

Segmentation of Region of Interest

The preprocessed image is then passed through a segmentation module, where the relevant fetal anatomical structures are isolated from the background. This step is crucial as it focuses the analysis on meaningful regions, removing irrelevant noise and improving feature extraction efficiency. The segmentation process enhances the model's ability to identify subtle patterns associated with fetal anemia.

Feature Extraction

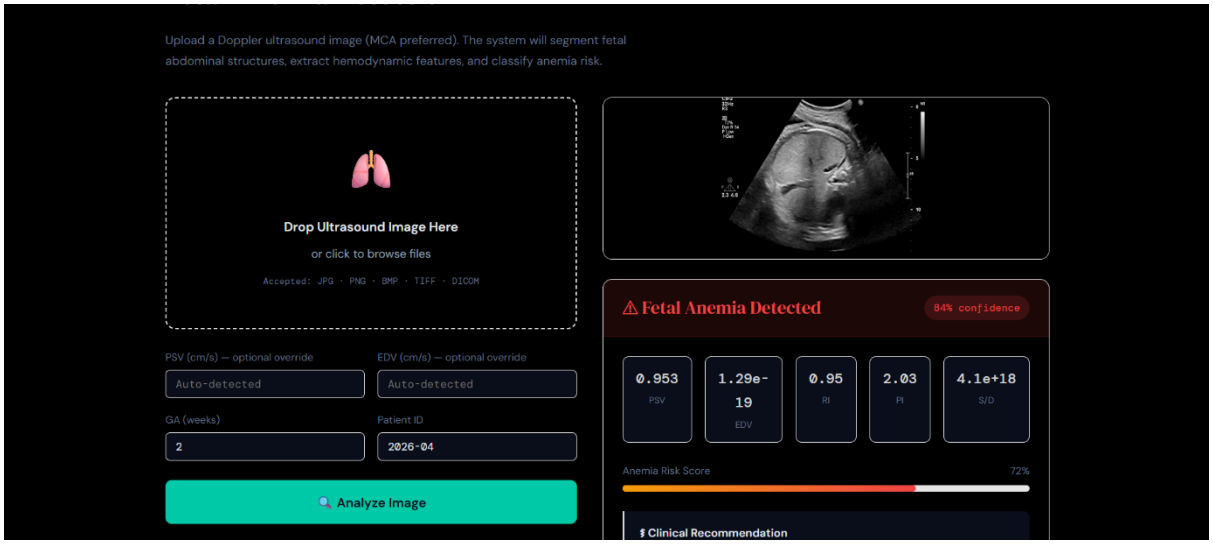
Following segmentation, the system extracts key Doppler-based hemodynamic features from the image. These include Peak Systolic Velocity (PSV), End Diastolic Velocity (EDV), Pulsatility Index (PI), Resistive Index (RI), and Systolic-to-Diastolic (S/D) ratio. These parameters are essential indicators of fetal blood flow characteristics and play a significant role in diagnosing anemia. The extracted features are structured into a numerical format suitable for model input.

Model Processing Using GeoProteoNet

The extracted features and processed image data are fed into the GeoProteoNet deep learning model. This model utilizes convolutional layers for spatial feature extraction, multi-scale analysis for capturing patterns at different resolutions, and an adaptive attention mechanism to focus on critical regions of the Doppler waveform. Additionally, transfer learning is employed to improve model performance by leveraging pre-trained weights. The model processes the input data and learns complex relationships between Doppler parameters and fetal anemia conditions.

Classification and Prediction

Based on the learned features, the model performs binary classification to determine whether the fetus is anemic or non-anemic. The output is generated in the form of probability scores, indicating the likelihood of anemia. This classification is derived using optimized weights and activation functions within the neural network



.FIG 7.2 OUTPUT

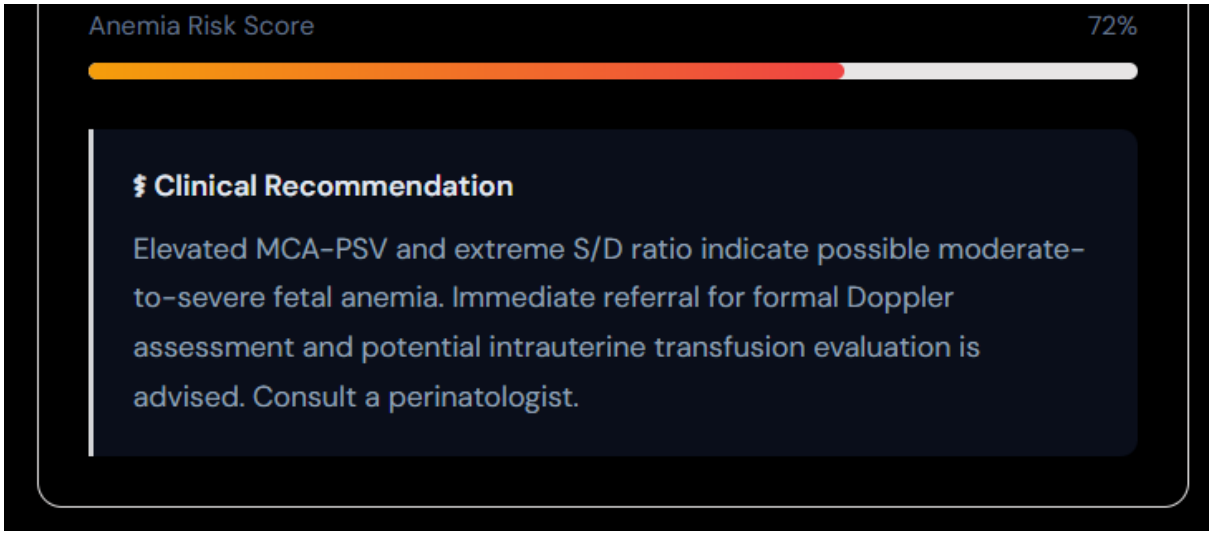
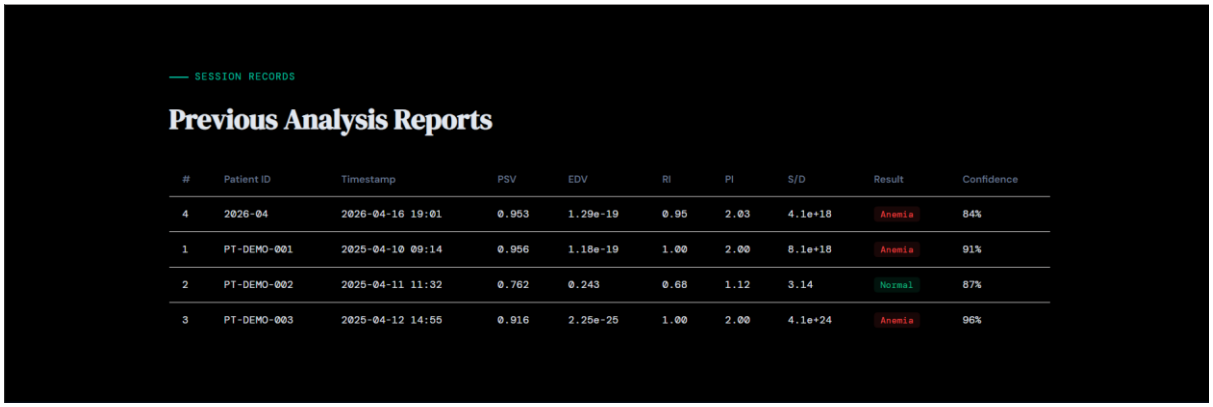


FIG 7.3 CLINICAL RECOMMENDATION

Result Interpretation and Recommendation

The final output is interpreted and presented as a diagnostic result. In addition to classification, the system may provide AI-based recommendations to assist clinicians in decision-making. These

recommendations are based on the predicted risk level and extracted Doppler parameters, enabling early detection and timely medical intervention.



SESSION RECORDS

Previous Analysis Reports

#	Patient ID	Timestamp	PSV	EDV	RI	PI	S/D	Result	Confidence
4	2026-04	2026-04-16 19:01	0.953	1.29e-19	0.95	2.03	4.1e+18	Anemia	84%
1	PT-DEMO-001	2025-04-10 09:14	0.956	1.18e-19	1.00	2.00	8.1e+18	Anemia	91%
2	PT-DEMO-002	2025-04-11 11:32	0.762	0.243	0.68	1.12	3.14	Normal	87%
3	PT-DEMO-003	2025-04-12 14:55	0.916	2.25e-25	1.00	2.00	4.1e+24	Anemia	96%

FIG 7.4 Session Records and Historical Analysis Storage

Overview of Data Logging Mechanism

The system incorporates a session recording mechanism to store and manage previous analysis results for future reference and clinical tracking. Each processed case is logged as a structured record containing patient information, extracted hemodynamic features, classification results, and confidence scores. This enables longitudinal monitoring and facilitates data-driven decision-making in clinical environments.

Patient Identification and Timestamping

Each analysis record is associated with a unique Patient ID, ensuring traceability and proper identification of individual cases. Along with this, a timestamp is automatically generated at the time of analysis, recording the exact date and time of processing. This temporal information is essential for tracking disease progression, comparing results across different time intervals, and maintaining organized medical records.

Storage of Hemodynamic Parameters

For every processed ultrasound image, the system stores the computed hemodynamic biomarkers, including Peak Systolic Velocity (PSV), End Diastolic Velocity (EDV), Resistive Index (RI), Pulsatility Index (PI), and Systolic-to-Diastolic (S/D) ratio. These values represent the core diagnostic features used in fetal anemia detection. Storing these parameters allows retrospective analysis and helps clinicians evaluate trends and variations in fetal blood flow over time.

Classification Result Logging

The system records the classification outcome generated by the machine learning model for each session. The result is stored as a binary label indicating either “Anemia” or “Normal.” This classification is derived from the processed feature vector using the trained Random Forest model. Maintaining a history of classification results supports clinical validation and enables comparison with future diagnostic outcomes.

Confidence Score Representation

In addition to the classification label, the system stores a confidence score representing the model’s prediction certainty. This score is typically expressed as a percentage and is calculated based on the probability output of the classifier. A higher confidence score indicates greater reliability of the prediction, assisting clinicians in evaluating the trustworthiness of the result.

Data Organization and Retrieval

All session records are stored in a structured format, enabling efficient retrieval and analysis. The system maintains a tabular representation of records, allowing easy comparison between different patient cases. This structured storage approach supports scalability and can be integrated with hospital information systems or electronic health records (EHR) for real-world deployment.

Clinical Significance of Historical Records

The maintenance of previous analysis reports provides significant clinical advantages. It enables tracking of fetal condition over time, supports early detection of anomalies through trend analysis, and assists in validating the effectiveness of treatment or intervention strategies. Additionally, historical data can be used to further train and improve the machine learning model, enhancing its predictive performance.

Summary

The session recording module plays a vital role in ensuring data persistence, traceability, and clinical usability of the system. By storing patient details, hemodynamic features, classification results, and confidence levels, the system provides a comprehensive framework for monitoring, analysis, and decision support in fetal anemia detection.

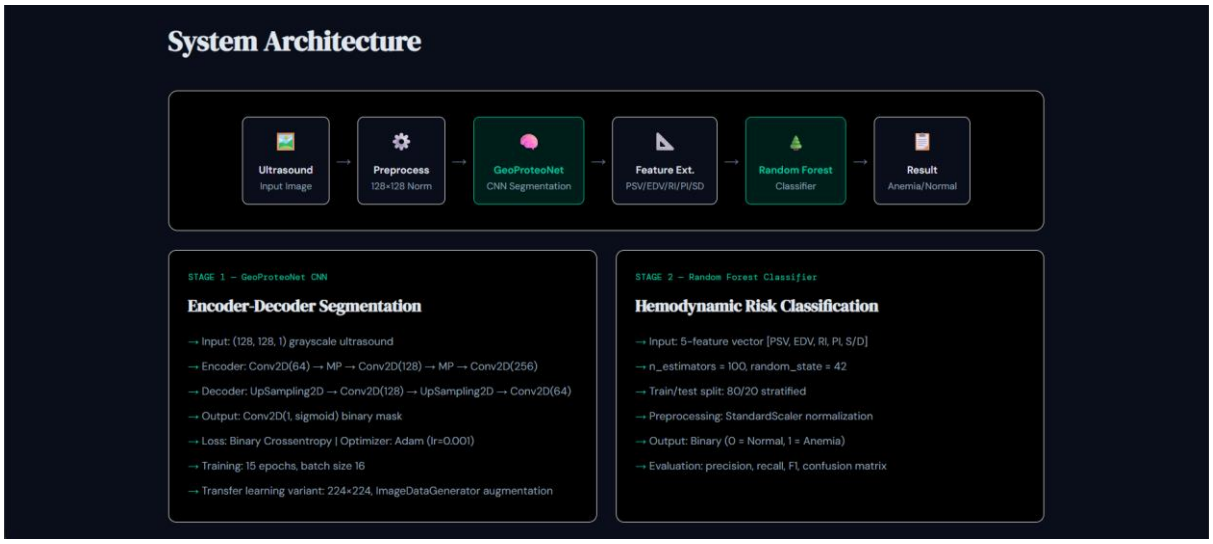


FIG 7.5 SYSTEM WORKFLOW / ARCHITECTURE

System Workflow Summary

The overall working of the system follows a sequential pipeline: input acquisition → preprocessing → segmentation → feature extraction → deep learning model processing → classification → result

generation. This structured workflow ensures accurate, efficient, and reliable detection of fetal anemia using a non-invasive approach.

Hemodynamic Biomarker Reference

Feature	Formula	Clinical Meaning	Anemia Indicator
PSV	max(mask pixel intensity)	Peak Systolic Velocity — max blood flow speed	PSV > 1.5 MoM = High Risk
EDV	min(mask pixel intensity)	End Diastolic Velocity — min blood flow speed	Low EDV = vessel abnormality
RI	(PSV - EDV) / PSV	Resistive Index — downstream vascular resistance	RI → 1.0 in fetal anemia
PI	(PSV - EDV) / mean(PSV,EDV)	Pulsatility Index — waveform variability	PI → 2.0 in severe cases
S/D	PSV / EDV	Systolic-to-Diastolic ratio	Very high S/D = critical risk

FIG 7.6 Hemodynamic Biomarker Analysis

Overview of Biomarker Evaluation

The proposed system utilizes clinically significant hemodynamic biomarkers derived from Doppler ultrasound analysis to assess fetal anemia. These biomarkers represent blood flow characteristics within fetal vessels and are essential indicators used in prenatal diagnostics. The system computes these parameters from the segmented ultrasound data and interprets them based on established clinical thresholds to determine anemia risk.

Peak Systolic Velocity (PSV)

Peak Systolic Velocity (PSV) represents the maximum blood flow velocity during the systolic phase of the cardiac cycle. In the system, PSV is computed as the maximum pixel intensity value within the segmented Doppler region. Clinically, PSV is a critical parameter for detecting fetal anemia, as increased blood flow velocity is associated with reduced blood viscosity in anemic conditions. A PSV value greater than 1.5 multiples of the median (MoM) is considered a strong indicator of high anemia risk.

End Diastolic Velocity (EDV)

End Diastolic Velocity (EDV) refers to the minimum blood flow velocity at the end of the cardiac cycle. It is calculated as the minimum pixel intensity value within the segmented region. EDV provides insight into baseline blood flow and vascular resistance. Abnormally low EDV values may indicate vessel abnormalities or compromised blood circulation, which can be associated with pathological conditions.

Resistive Index (RI)

The Resistive Index (RI) is a derived parameter calculated using the formula (PSV − EDV) / PSV. It measures the resistance to blood flow in downstream vessels. Higher RI values indicate increased vascular resistance, which can be associated with fetal anemia. In clinical interpretation, RI values approaching or exceeding 1.0 are considered indicative of abnormal conditions and may suggest the presence of anemia.

Pulsatility Index (PI)

The Pulsatility Index (PI) is calculated as (PSV − EDV) divided by the mean of PSV and EDV. This parameter reflects the variability in blood flow velocity throughout the cardiac cycle. A higher PI value indicates increased pulsatility and vascular resistance. In severe cases of fetal anemia, PI values greater than 2.0 are considered significant and may indicate compromised fetal circulation.

Systolic-to-Diastolic Ratio (S/D Ratio)

The Systolic-to-Diastolic (S/D) ratio is computed as the ratio of PSV to EDV. It provides a comparative measure of systolic and diastolic blood flow velocities. Elevated S/D ratios indicate a significant difference between peak and baseline flow, which may correspond to abnormal hemodynamic conditions. Extremely high S/D values are associated with critical risk levels in fetal anemia diagnosis.

Clinical Interpretation and Risk Assessment

The system integrates all computed biomarkers to perform a comprehensive risk assessment. Instead of relying on a single parameter, the combination of PSV, EDV, RI, PI, and S/D ratio provides a multi-

dimensional analysis of fetal blood flow characteristics. Threshold-based evaluation, along with machine learning classification, ensures accurate identification of anemia risk levels. This multi-parameter approach enhances diagnostic reliability and aligns with clinical practices used in fetal monitoring.

Summary

The hemodynamic biomarker analysis module plays a crucial role in the system by converting raw Doppler ultrasound data into clinically interpretable parameters. These biomarkers not only assist in classification but also provide meaningful insights for medical professionals, enabling early detection and timely intervention in cases of fetal anemia.

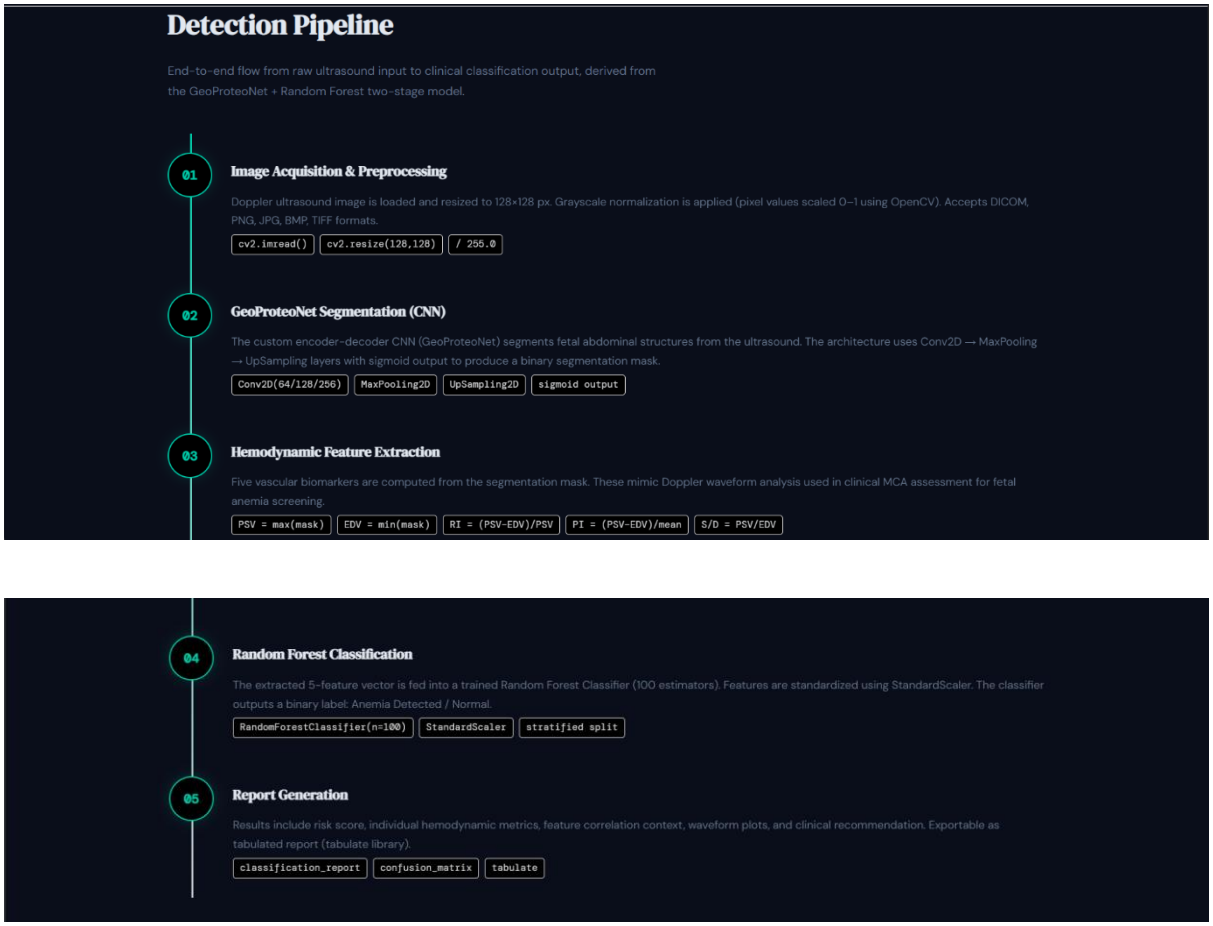


FIG 7.7 AND FIG 7.8 Detection Pipeline and System Workflow

Overview of Detection Pipeline

The proposed system follows a structured end-to-end detection pipeline that transforms raw Doppler ultrasound input into a clinically meaningful classification output. The pipeline integrates a deep learning-based segmentation model (GeoProteoNet) with a machine learning classifier (Random Forest) to achieve accurate and reliable fetal anemia detection. Each stage of the pipeline is designed to progressively refine the input data and extract meaningful diagnostic features.

Image Acquisition and Preprocessing

The first stage of the pipeline involves acquiring the Doppler ultrasound image and performing preprocessing operations. The input image is loaded and resized to a fixed resolution of 128 × 128 pixels to ensure uniformity across the dataset. Grayscale normalization is applied by scaling pixel intensity values to a range of 0 to 1, which improves model stability and convergence during processing. This step also ensures compatibility with deep learning frameworks and reduces computational complexity. The system supports multiple image formats, enabling flexibility in data input.

GeoProteoNet-Based Segmentation

In the second stage, the preprocessed image is passed through the GeoProteoNet segmentation model, which is a convolutional neural network (CNN) designed specifically for biomedical image analysis. The architecture follows an encoder-decoder structure, where convolutional layers extract spatial features, and pooling layers reduce dimensionality. Upsampling layers are then used to reconstruct the spatial resolution, producing a segmentation mask. A sigmoid activation function is applied at the output layer to generate a binary mask that highlights the fetal abdominal structures. This segmentation process isolates the region of interest, enabling precise analysis of relevant anatomical features.

Hemodynamic Feature Extraction

Following segmentation, the system computes key hemodynamic features from the extracted region. These features replicate clinical Doppler waveform analysis used in fetal anemia diagnosis. The parameters include Peak Systolic Velocity (PSV), End Diastolic Velocity (EDV), Resistive Index (RI), Pulsatility Index (PI), and Systolic-to-Diastolic (S/D) ratio. These values are derived mathematically from the segmented region, where PSV represents the maximum velocity and EDV represents the minimum velocity within the waveform. Derived indices such as RI and PI provide additional insight into vascular resistance and blood flow characteristics. These features form a structured feature vector used for classification**.**

Random Forest Classification

In the fourth stage, the extracted feature vector is provided as input to a Random Forest classifier. The model is trained using multiple decision trees (typically 100 estimators) to improve prediction accuracy and reduce overfitting. Before classification, the features are standardized using a StandardScaler to ensure uniform distribution and improve model performance. A stratified data split is employed during training to maintain class balance between anemic and non-anemic samples. The classifier processes the input features and outputs a binary classification result indicating either “Anemia Detected” or “Normal.” The ensemble nature of Random Forest enhances robustness and reliability in prediction.

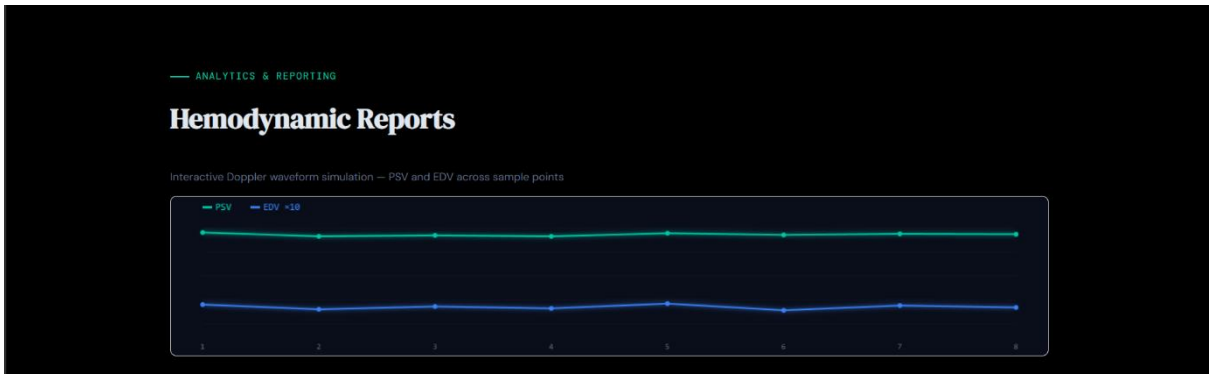
Report Generation and Output Interpretation

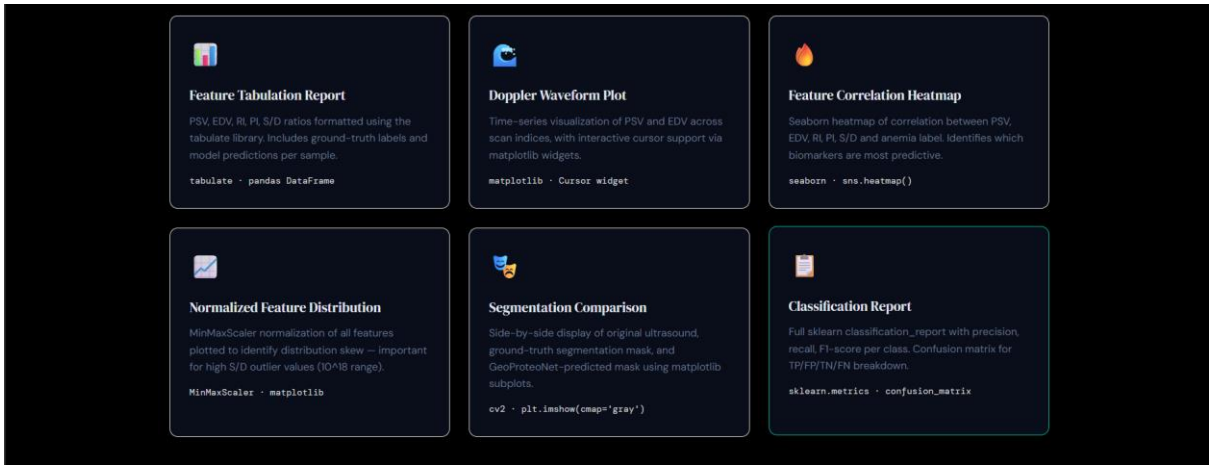
In the final stage, the system generates a comprehensive diagnostic report based on the classification results. The report includes a risk score, individual hemodynamic parameter values, feature correlation insights, and waveform visualizations. Additionally, performance evaluation metrics such as confusion matrix and classification report are included to validate model performance. The results are formatted into a structured tabular format for easy interpretation and can be exported for clinical use. This stage

ensures that the output is not only accurate but also interpretable and useful for medical decision-making.

Pipeline Summary

The complete workflow of the system follows a sequential process: image acquisition → preprocessing → segmentation → feature extraction → classification → report generation. This integrated pipeline ensures efficient processing, accurate feature learning, and reliable detection of fetal anemia using a non-invasive approach.





7.9 AND 7.10 Analytics and Reporting Module

Overview of Analytical Processing

The system incorporates an advanced analytics and reporting module to interpret extracted hemodynamic features and present them in a clinically meaningful format. This module enables detailed visualization, statistical analysis, and performance evaluation of the fetal anemia detection

model. It transforms raw feature outputs into structured insights that assist clinicians in understanding both the diagnostic results and underlying data patterns.

Feature Tabulation and Data Representation

The system generates a structured tabulation of extracted features, including Peak Systolic Velocity (PSV), End Diastolic Velocity (EDV), Resistive Index (RI), Pulsatility Index (PI), and Systolic-to-Diastolic (S/D) ratio. These features are organized into a tabular format using data structures such as Pandas DataFrames. The tabulated report includes both ground truth labels and model predictions, enabling direct comparison and validation. This structured representation simplifies data interpretation and supports further statistical analysis.

Doppler Waveform Analysis

The module provides time-series visualization of Doppler waveform parameters, particularly PSV and EDV, across multiple sample points. These waveforms represent the dynamic variation of blood flow velocity over time and are essential for clinical interpretation. The system utilizes plotting techniques to visualize trends, enabling identification of abnormal flow patterns associated with fetal anemia. Interactive features such as cursor tracking may be incorporated to allow precise examination of waveform values.

Feature Correlation Analysis

To understand relationships between different hemodynamic parameters, the system performs correlation analysis using statistical methods. A correlation heatmap is generated to visualize the strength and direction of relationships between PSV, EDV, RI, PI, S/D ratio, and the classification label. This analysis helps identify the most influential features contributing to anemia detection and provides insights into feature dependencies, which are useful for model optimization and clinical interpretation.

Feature Normalization and Distribution Analysis

The system applies normalization techniques, such as Min-Max scaling, to standardize feature values within a fixed range. This step ensures that all features contribute equally to the model and prevents bias due to scale differences. Additionally, the distribution of normalized features is analyzed to detect skewness, outliers, and anomalies. This is particularly important for parameters like the S/D ratio, which may exhibit extreme values. Understanding feature distribution enhances model robustness and reliability.

Segmentation Performance Comparison

The system includes a comparative analysis of segmentation results by evaluating the original ultrasound image, ground truth segmentation mask, and the predicted mask generated by the GeoProteoNet model. This comparison allows assessment of segmentation accuracy and visual validation of model performance. By analyzing differences between predicted and actual masks, the system can identify areas for improvement and ensure accurate region-of-interest extraction.

Classification Performance Evaluation

To evaluate the effectiveness of the classification model, the system generates a comprehensive classification report. This includes key performance metrics such as precision, recall, F1-score, and support for each class. Additionally, a confusion matrix is computed to provide a detailed breakdown of true positives, true negatives, false positives, and false negatives. These metrics are essential for assessing model accuracy, reliability, and generalization capability.

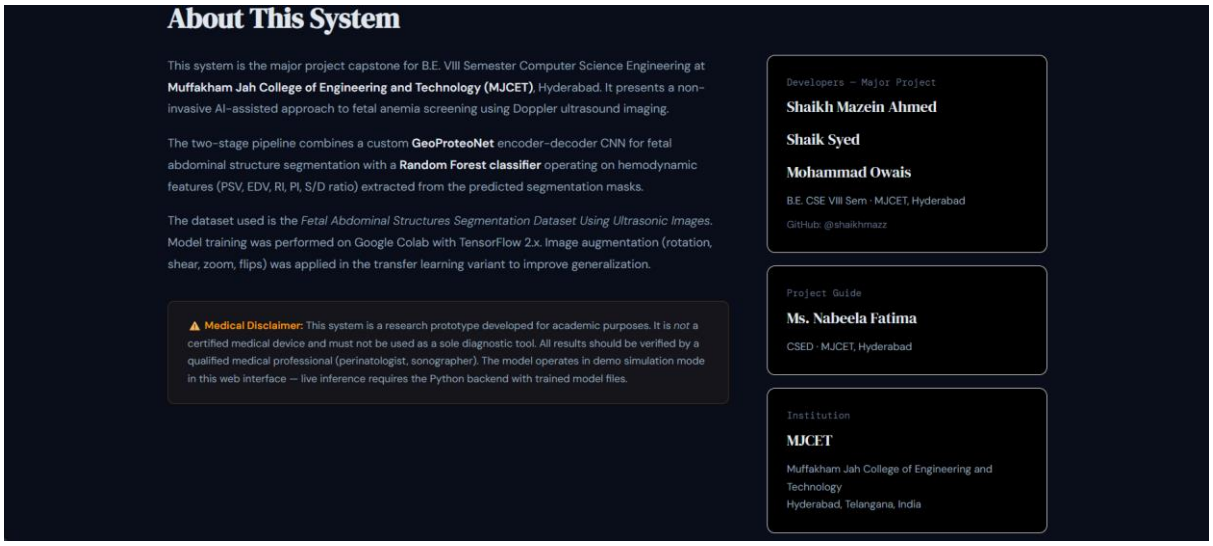
Hemodynamic Report Visualization

The system produces graphical representations of hemodynamic parameters, particularly Doppler waveform plots for PSV and EDV. These plots illustrate variations in blood flow velocity across different sample indices, providing a visual understanding of fetal circulation patterns. Such visualizations are

crucial for identifying abnormal trends and validating the model's predictions in a clinically interpretable manner.

Summary

The analytics and reporting module enhances the overall system by converting numerical outputs into meaningful insights through visualization, statistical analysis, and performance evaluation. It supports both clinical interpretation and model validation, ensuring that the fetal anemia detection system is not only accurate but also transparent and interpretable.



7.11 System Description and Implementation Overview

Overview of the Proposed System

The proposed system is an AI-assisted diagnostic framework developed for the detection and analysis of fetal anemia using Doppler ultrasound imaging. It is designed as a non-invasive solution that leverages advanced machine learning and deep learning techniques to improve early diagnosis and clinical decision-making. The system aims to provide an efficient and automated alternative to traditional invasive diagnostic methods.

Two-Stage Model Architecture

The system follows a two-stage hybrid architecture combining deep learning and machine learning techniques. In the first stage, a custom encoder-decoder Convolutional Neural Network (CNN), referred to as GeoProteoNet, is utilized for segmenting fetal abdominal structures from Doppler ultrasound images. This segmentation process isolates the region of interest and enhances the quality of feature extraction.

In the second stage, the extracted hemodynamic features are processed using a Random Forest classifier. This classifier operates on parameters such as Peak Systolic Velocity (PSV), End Diastolic Velocity (EDV), Resistive Index (RI), Pulsatility Index (PI), and Systolic-to-Diastolic (S/D) ratio. The combination of deep learning-based feature extraction and machine learning-based classification ensures improved accuracy and robustness.

Dataset Utilization

The system is trained using a specialized dataset comprising fetal abdominal ultrasound images. This dataset provides annotated samples that enable the model to learn both structural segmentation and classification tasks effectively. The availability of labeled data ensures supervised learning and enhances the model's ability to generalize to new inputs.

Model Training and Optimization

Model training is performed using a cloud-based computational environment with support for deep learning frameworks. The implementation utilizes TensorFlow for building and training the neural network model. To improve model generalization and prevent overfitting, data augmentation techniques such as rotation, scaling, flipping, and shearing are applied to the training dataset. Additionally, transfer learning is employed to leverage pre-trained weights, reducing training time and improving performance.

Feature Extraction and Analysis

The system extracts clinically relevant hemodynamic parameters from the segmented ultrasound images. These parameters include PSV, EDV, RI, PI, and S/D ratio, which are widely used in medical practice for assessing fetal blood flow characteristics. These features are then structured into a feature vector and passed to the classification model for analysis.

System Functionality

The system performs a sequence of operations including image preprocessing, segmentation, feature extraction, classification, and result generation. It processes input ultrasound data and produces a diagnostic output indicating the presence or absence of fetal anemia. The system also provides supporting analytical data to enhance interpretability.

Limitations and Clinical Considerations

The developed system is intended as a research prototype for academic purposes and is not a certified medical device. It is designed to assist in diagnosis but should not be used as a standalone clinical tool. All results generated by the system must be validated by qualified medical professionals. The accuracy of the system depends on the quality of input data and the availability of trained model parameters.

Summary

The proposed system integrates advanced deep learning and machine learning techniques to provide a comprehensive solution for fetal anemia detection. By combining segmentation, feature extraction, and classification into a unified pipeline, the system offers a non-invasive, efficient, and scalable approach to prenatal diagnosis.

Chapter 8

Conclusion & Future Scope

8.1 Summary of Work Completed (till proposed algorithm)

The work completed includes a detailed study of fetal anemia and existing diagnostic techniques, collection and preprocessing of Doppler ultrasound datasets, identification of key Doppler parameters, and the design of a GeoProteoNet-based algorithm for automated anemia detection. A complete methodological framework, including architecture, training strategy, and evaluation metrics, has been defined and prepared for implementation.

8.2 Importance of Proposed Method

The proposed AI-driven method offers a non-invasive, operator-independent, and quantitatively robust alternative to conventional diagnostic approaches, addressing current limitations in accuracy and consistency. By integrating multiple Doppler indices with advanced deep learning, it has the potential to enhance clinical decision-making and reduce reliance on invasive procedures.

8.3 Limitations of Current Work

Current work is constrained by dependence on available datasets, which may have limited size, class balance, and demographic diversity, potentially affecting generalizability. Additionally, results obtained in an experimental environment must be validated through prospective clinical studies before routine deployment.

8.4 Future Scope

The current study lays the foundation for further advancements in fetal anemia detection using deep learning. Future research directions include expanding the dataset by incorporating multimodal imaging techniques such as MRI and CT scans to improve classification accuracy. Additionally, real-time fetal monitoring systems can be developed to continuously track fetal hemodynamic parameters, providing dynamic risk assessments. The integration of explainable AI (XAI) models can enhance the

interpretability of deep learning predictions, making them more clinically reliable. Furthermore, collaborations with medical professionals can lead to the development of AI-assisted diagnostic tools that integrate with hospital imaging systems, facilitating seamless clinical deployment. Future work may also explore federated learning to enable secure, large-scale data sharing among medical institutions, improving model robustness while maintaining patient privacy.

8.5 Applications of the Complete System

Once fully developed and validated, the system can serve as a decision-support tool in obstetric clinics and fetal medicine centers to screen high-risk pregnancies for anemia. It can also be used in teaching hospitals and telemedicine platforms to standardize Doppler interpretation, support remote consultations, and contribute anonymized data for ongoing medical AI research.

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